

A Major Project Final Report on

VeriFact: Verification of Health Information

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ABSTRACT

This project aims to help people in Nepal easily identify and understand health misinformation that spreads across social media. False claims about medicines, diseases, home remedies, treatments or public health updates can circulate quickly and harm individuals who rely on them. To address this issue, we are creating a system that can automatically detect misleading or false health information. The system analyzes both text and images. If a user uploads a photo or screenshot, the OCR module extracts the text, which is then processed by our AI model to determine whether the content appears trustworthy, misleading or completely false. The project is being developed in stages, including dataset collection, AI model training, mobile app development, OCR integration and backend implementation using FastAPI. Throughout the process, we store analysis results, user feedback and verified facts in a database so the system can improve over time and provide insights such as trending misinformation topics and frequently observed false sources. Ultimately, our goal is to give people a simple and reliable tool to check suspicious health content, understand its background and make safer decisions. Despite many challenges, we hope this system will help reduce the impact of health misinformation and support more informed communities.

Keywords: Health misinformation, Social media, False health claims, OCR, Misinformation detection, FastAPI backend, Image and text analysis, Verified health sources, User feedback

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1 INTRODUCTION

These days, the digital world is enriched with information and misinformation, given the massive amount of unregulated user generated content on the internet, especially on social networks. "Misinformation" is defined as information that is untrue, inaccurate, incomplete, lacking evidence, intended to deceive or demonstrably false. The rapid expansion of social media platforms has transformed the way people access and share health related information.

Although these platforms make information easily accessible, they also allow false or misleading health content to spread very quickly [14]. Such misinformation can confuse people, promote unsafe practices, delay proper treatment and even reduce trust in healthcare professionals. This widespread mix of true and false information is often referred to as an "infodemic". [7].

Research shows that health misinformation is common across many online platforms. Health misinformation on social media can be seen in multiple forms that can range from exaggerated claims about treatment to false narratives regarding different diseases [22]. These studies highlight the urgent need for automated systems that are capable of detecting misleading content. Because of this growing problem, many researchers have studied different technological approaches to automatically detect and reduce health misinformation [14].

Checking every post manually is not possible because social media produces a huge amount of content every day. Therefore, automated systems using artificial intelligence and machine learning are needed. Early studies used crowdsourcing and simple classifiers to identify false medical claims on Twitter [11]. More recent studies use deep learning methods that can better understand language patterns. Some models also combine text and images, since many misleading posts use both forms of content [25].

Recent advancements in AI have provided tools to tackle misinformation. These systems analyze both text and images to spot false content, helping to identify misleading posts quickly [25]. These methods use advanced AI techniques to understand and analyze content that allows the system to identify misinformation.

Studies emphasize the need for approaches that not only identify false content but also understand its impact [5]. By automatically identifying false health information, this system will help provide reliable guidance to users, particularly those who are more likely to be influenced by misinformation [7, 24]. Hence, this project aims to develop a system that can detect health misinformation online.

1.1 Problem Statement

Social media makes it very easy for people to access health information. Many individuals search online for advice about diseases, medicines and treatments. However, it also enables the rapid spread of incorrect or misleading health information. Many people especially older adults tend to believe what they read online without verifying its authenticity. This can lead to poor health decisions such as trying unproven treatments or ignoring proper medical advice.

Health misinformation can encourage unsafe treatments or home remedies, delay proper medical care, increase fear and panic among the public and create an infodemic where an overwhelming amount of information makes it difficult to distinguish accurate facts from false claims [7]. Research further indicates that a significant portion of health related posts shared online contain incorrect or low quality information [22].

Manually checking each post is not practical because millions of posts are generated daily, human experts do not have the capacity to verify every piece of information and false information often spreads faster than verified facts. Due to these challenges, there is a strong need to develop an AI based system that can verify whether a health claim is accurate or misleading, better understand medical language and provide trustworthy and reliable information to users.

Therefore, this project aims to address this issue by developing a system capable of analyzing health related posts in both image and textual formats, comparing them against reliable medical sources and determining whether the claim is supported or refuted along with a confidence score.

1.2 Project Objectives

The main goal of this project is to build a system that helps people to identify which health information is reliable or unreliable. The system will make it easier for users to understand if the claim is reliable. The objectives of this project are:

- Detect health related misinformation in social media posts, news
- To develop an AI based model, such as BioBERT to classify posts as supported, refuted or neutral with confidence score

1.3 Significance of the Study

Health misinformation spreads quickly across online platforms reaching people who trust what they read without verifying it. Understanding the importance of detecting, such misinformation is crucial for all individuals(users). The following points highlight the significance of the study:

- **Helps to identify false health information:** The system enables users to detect misleading or incorrect health claims on online platforms that helps them avoid being misinformed.
- **Supports reliable health decisions:** By distinguishing accurate information from false claims, users can make safer and more informed choices about their health.
- **Promotes informed decision making:** With verified evidence, users can make safer choices about health advice or medicines.
- **Contributes to academic research:** Demonstrates the use of AI and biomedical language models, like BioBERT, in detecting domain-specific health misinformation.
- **Reduces personal risk from misinformation:** By limiting exposure to false claims, the system protects users from potentially harmful advice.
- **Promotes understanding of online health content:** Helps in analyzing patterns in how misleading posts are written or presented.

1.4 Scope and Limitation

Scope:

This project focuses on verifying health misinformation shared across social media posts. It utilizes a curated set of verified datasets for claim verification and primarily targets common health topics such as diseases, treatments and preventive measures. The system employs artificial intelligence techniques along with biomedical language models such as BioBERT to analyze textual content effectively. Additionally, it integrates AI and OCR based detection to process both image and text data. The results are delivered through a mobile application interface to enhance accessibility and user interaction. The performance of the model is evaluated using standard metrics, including accuracy, precision, recall and F1-score.

Limitation:

Despite its capabilities, the system has certain limitations. The accuracy of the model depends heavily on the availability and quality of reliable evidence sources online. It is limited by the size and diversity of the dataset which may not cover all types of health misinformation. Newly emerging or previously unseen health misinformation that is not included in the dataset may reduce detection accuracy. Furthermore, the system cannot verify claims in languages that are not supported by its translation module. Currently, results are provided only in English which may limit usability for users who are not comfortable with the language. Additionally, the system does not provide medical advice. It only informs users about the reliability of the health information presented.

2 LITERATURE REVIEW

The rapid growth of social media and digital communication platforms has transformed how health information is consumed. While these platforms enable fast knowledge sharing and community support, they also facilitate the spread of health misinformation. This leads to harmful behaviors, delayed treatments and reduced trust in healthcare systems. Health misinformation has emerged as a major global concern, prompting extensive research on its patterns, effects and technological solutions. Understanding the magnitude of the problem is essential alongside developing technical detection methods.

2.1 Dynamics and Pattern of Health Misinformation

Studies have shown that misinformation spreads rapidly due to user engagement dynamics, echo chambers and algorithmic amplification of deceptive content [14]. Technological solutions have been categorized into machine learning classification, deep learning models and knowledge based verification systems. Challenges include limited labeled datasets, evolving misinformation patterns and insufficient biomedical expertise suggesting that specialized language models can better capture medical semantics than generic NLP models. It has also been confirmed that misinformation is prevalent across major platforms, particularly around vaccines, diets, pandemics and chronic illnesses. Emotional narratives and sensational structuring contribute to misinformation spreading faster than verified information which highlights the need for scalable automated detection systems [22].

2.2 Advances in Transformer Based Detection Models

Recent developments in transformer based models have improved text understanding and misinformation detection. For example, BERT based multimodal models combining text and images emphasize the importance of context-aware representations [2]. However, the use of general domain BERT rather than biomedical variants limits effectiveness for specialized health terminology, suggesting domain adapted models like BioBERT may improve performance. Models integrating socio contextual features such as user interactions and post meta-data demonstrates that combining textual and contextual signals improves classification performance [24]. Yet, reliance on general domain embeddings still leaves room for biomedical pretraining.

2.3 Evolutionary and Feature Based Approaches

Evolutionary algorithms have been proposed for COVID-19 misinformation detection using feature selection and heuristic optimization to improve performance [1]. While effective, such approaches heavily depend on handcrafted features and may not scale as well as transformer based architectures. Early computational approaches focused mainly on text based machine learning but the complexity of misinformation is driving the adoption of multimodal techniques. Combining text and image cues in deep learning models has been shown to significantly improve medical misinformation detection [25]. However, these methods do not incorporate domain specific biomedical language representations which suggests opportunities to integrate pretrained biomedical models with multimodal architectures.

2.4 Social and Public Health Perspectives

Beyond computational methods, social strategies are essential for combating misinformation. Combining authoritative health content with community engagement reduces the spread of false information [17]. From a global perspective, infodemics are often equated with biological epidemics which emphasizes the need for multi-sectoral strategies integrating regulation, technology and public education [7]. Governments and health organizations are increasingly adopting AI based tools such as automated fact checking, chatbots and content moderation systems, while technical solutions alone are insufficient without reliable classification accuracy and ethical deployment [6].

2.5 Linguistic Indicators and Semantic Cues

Finally, studies on linguistic characteristics provide additional insights into detection. Posts with false information such as untested cancer treatments often employ emotional language, anecdotal evidence and persuasive claims rather than scientific terminology [9]. Readability, certainty markers and subtle semantic cues can accurately distinguish deceptive content. These findings indicate that models capable of capturing nuanced semantics are crucial for reliable health misinformation detection.

3 METHODOLOGY

3.1 Software Development Life Cycle

This project follows the **Software Development Life Cycle (SDLC)** using the **Incremental Model** allowing the system to be developed in fully functional modules. Each increment adds new capabilities to the system. The whole development process is divided into six increments.

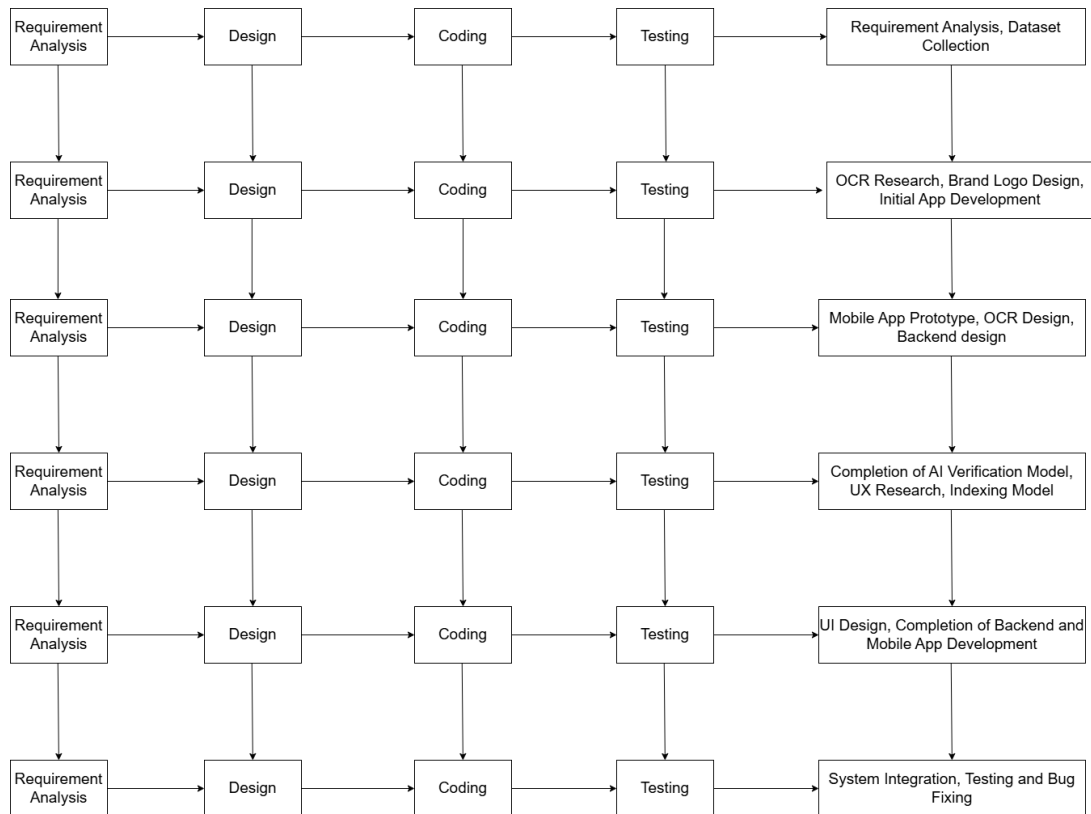


Figure 1: Incremental Model

1. Increment 1: Requirement Analysis & Dataset Collection

- Conducted requirement gathering to understand user needs and defined scope of the system
- Collected datasets for training and evaluating the AI models
- Explored different machine learning and transformer-based approaches
- Prepared technical documentation

2. Increment 2: OCR Research & Brand Logo Design & Initial Mobile App Development

- Researched and evaluated OCR techniques for extracting text from images
- Began mobile application development (OCR plugin Integration)
- Crafted the app logo to strengthen brand identity and visual impact
- Continued expanding the dataset for better model training.
- Conducted a comparative study of various NLP models for health claim verification
- Documentation (Ongoing)

3. Increment 3: Mobile App Development& Backend Setup

- Development a working prototype of the mobile application with basic features
- Setup Backend technologies and began designing the backend system
- Implemented data harvesting and scraping modules to collect health evidence from trusted sources
- Identified Bio- BERT with LoRA (Low-Rank Adaptation) as the most efficient and accurate approach
- Documentation (ongoing)

4. Increment 4: AI Verification Model & UX Research & Indexing Model

- Finalized AI model to verify health information
- Developed an evidence indexing model to generate BioBERT embeddings
- Began in-depth UX Research covering user personas, user journey maps and value propositions

- Documentation (ongoing)

5. Increment 5: UI Design & Completion of Backend and Mobile App Development

- Designed detailed UI/UX prototypes in Figma, including navigation, themes and interaction flow
- Finalized the health claim verification model.
- Completed backend development with database integration for storing user queries and history
- Finalized core mobile app functionalities
- Prepared system diagrams and documentation (ongoing)

6. Increment 6: System Integration & Testing and Bug Fixing

- Integrated the mobile application, backend server, database, AI model and indexing modules into a unified system
- Enabled automated flow from user input (OCR/text) to preprocessing, prediction and result display
- Conducted end-to-end system testing and performance evaluation
- Identified and resolved bugs
- Prepared deployment-ready build
- Completed Documentation

3.2 Technologies Used

TECHNOLOGIES TO BE USED	SUBJECT
Google Colab	Model Development and Training
Python	Model Implementation and System Logic
Fast API	Backend Development
Flutter	Mobile Application Development
Figma	UI/UX Design and Prototyping
Hive	Backend and Database
Git and GitHub	Version Control
VS Code	Code Editor
LaTeX	Documentation
Pycharm	Code Editor
Notion	Project Management

Table 1: Technologies used

Justified for Selection

Google Colab : Google Colab is an online platform that allows coding and running Python in the cloud. It provides free access to GPUs which helps in training models efficiently. For our project, Colab is used to develop and train the health information verification model without requiring high-end hardware. Its collaborative features allow multiple team members to run experiments and share results easily.

Python : Python is a versatile programming language ideal for data processing and machine learning. In our project, it is used to implement the model, process health information and handle system logic. Its extensive libraries such as NumPy and PyTorch, simplifies model operations and integration. Python also works seamlessly with FastAPI and MongoDB for

backend tasks.

FastAPI : FastAPI is a modern Python framework to build fast and scalable APIs. For our project, it connects the trained model with the mobile app, providing verification results. Its asynchronous capabilities improve performance when handling multiple requests. FastAPI also generates API documentation automatically which simplifies testing and integration.

Flutter : Flutter is a framework for developing cross-platform mobile apps. For our project, Flutter ensures a responsive and user-friendly interface for submitting health information and receiving verification results. It integrates easily with backend APIs like FastAPI.

Figma : Figma is a design tool for creating UI/UX prototypes. In our project, it helps visualize the app layout, decide users types, plan user interactions and test usability before development. Its collaborative features allow team feedback and design refinement. Using Figma ensures the app is intuitive and visually appealing.

Hive : Hive is used in the backend as a data warehousing and query layer for managing and analyzing large scale structured datasets. It enables efficient storage and SQL based querying of distributed data making it suitable for handling high volume backend data processing and database operations.

Git and GitHub : Git is a version control system and GitHub hosts repositories online. They allow the team to collaborate, track changes and revert to previous code versions if needed. For our project, this ensures organized development and smooth teamwork.

VS Code : VS Code is a lightweight code editor with extensions for Python, Git and debugging. In our project, it helps write, test and debug code efficiently. Its features like syntax highlighting and integrated terminal improve productivity.

PyCharm : PyCharm is a Python IDE that provides advanced coding, debugging and project management tools. It is used in our project for writing clean and robust code, managing dependencies and integrating with Git. Its features simplify backend and model development.

LaTeX : LaTeX is a typesetting system for professional documentation. For our project, it is used to prepare structured reports, including tables, figures and formulas. It ensures consistency, readability and academic standards.

Notion : Notion is a project management tool that organizes tasks, notes and timelines. In our project, it helps track progress, assign responsibilities and collaborate efficiently. It ensures smooth planning and management of the project.

4 MODEL ARCHITECTURE AND IMPLEMENTATION

This section describes the detailed process of developing, training and evaluating our model for verifying the stance of evidence with respect to a claim. Each step has been carefully executed to ensure accuracy, generalization and efficiency of the model. The dataset we used is [19].

4.1 Dataset Overview and Preprocessing

The dataset used in this study is a curated health claim dataset consisting of claims, corresponding evidence and stance labels. Initially, the dataset contained 14,331 records with the following columns:

- `id`, `evidence`, `claim`, `label`, `topic_ip`, `question`

Data Cleaning and Preprocessing:

1. Removed header rows and missing values in `claim`, `evidence`, and `stance` columns.
2. Removed duplicate claim evidence pairs, resulting in a clean dataset of 13,309 records (92.9% retained).
3. Very short or garbled texts were removed to improve model training.
4. Conflicting queries (claims labeled both *supports* and *refutes*) were identified and removed to maintain data consistency, totaling 354 entries.
5. Text length features were computed for both `claim` and `evidence`, including character length and word count.

Label Distribution: The cleaned dataset contained the following stances:

- Neutral: 5,106 (38.4%)
- Supports: 4,982 (37.4%)
- Refutes: 3,221 (24.2%)

Dataset Splitting: Stratified splitting ensured class proportions were maintained:

- Training Set: 9,316 records (70%)
- Validation Set: 1,996 records (15%)
- Test Set: 1,997 records (15%)

All splits maintained the original stance distribution to avoid introducing bias.

4.2 Exploratory Data Analysis and Visualization

Exploratory Data Analysis (EDA) was conducted to understand textual characteristics, label distributions and potential class imbalances. Key insights include:

1. Text Length and Word Count:

- Average claim length: 108 characters (median 93)
- Average evidence length: 216 characters (median 183)
- Combined text length: mean 325 characters, maximum 1307

Histograms and box plots were generated to visualize the distribution of query and evidence lengths, revealing:

- Most queries are short, between 50 and 100 characters.
- Evidence texts are generally longer, peaking around 200 to 400 characters.

2. Word Frequency and Vocabulary Patterns:

- Word clouds were generated for *supports* and *refutes* claims, highlighting frequently occurring words.
- Common words helped reveal thematic differences between the stances.

3. Class Balance Analysis:

- The dataset is nearly balanced between *supports* (37.4%) and *refutes* (24.2%) stances with neutral claims (38.4%) forming a separate majority class.
- Confirms that class weighting may be applied during model training to handle imbalance effectively.

4. Dimensionality Reduction: Principal Component Analysis (PCA) was performed on BioBERT embeddings and other textual features to visualize separability:

- Two principal components captured a significant portion of variance.
- Visualizations showed partial separation of claims by stance, indicating that embeddings capture semantic differences.

4.3 Bio-BERT Based Classification Model

For text classification, we employed **BioBERT** (*dmis-lab/biobert-v1.1*), a BERT-based model pre-trained on large biomedical corpora which makes it suitable for health claim verification tasks.

1. Tokenization and Input Preparation:

- Texts were tokenized using Hugging Face’s `AutoTokenizer`.
- Maximum token length was set to 512, padding or truncating sequences accordingly.
- Tokenized sequences were converted into PyTorch tensors for model training.

2. Model Configuration:

- Number of classes: 3 (*supports, refutes, neutral*)
- LoRA (Low-Rank Adaptation) fine-tuning: Only 2.42% of model parameters are trainable which reduces computation while adapting to the new task.
- LoRA parameters: rank $r = 8$, alpha = 16, dropout = 0.15 on query, key, value and dense layers.

3. Performance: The BioBERT model achieved strong performance on the test set:

- Overall accuracy: 75.58%
- F1-score: 75.57%
- Precision/Recall by class:
 - Refutes: Precision 76.28%, Recall 77.51%, F1 76.89%
 - Supports: Precision 74.77%, Recall 73.45%, F1 74.11%

Calibration techniques were applied to reduce overconfidence in predictions:

- Only 5.2% of incorrect predictions had confidence > 0.9 .
- Brier score: 0.1596, indicating well-calibrated probability outputs.

4.4 Multi Feature Integration

To improve classification performance, we integrated additional textual and semantic features with BioBERT embeddings. This multi-feature approach leverages complementary information beyond the pre-trained model.

- **BioBERT embeddings:** 768 dimensional embeddings representing semantic meaning of claims and evidence.
- **Text length features:** Character count and word count of both claims and evidence.
- **Lexical features:** Number of unique words, stopwords, and punctuation marks.
- **Semantic similarity features:** Cosine similarity between claim and evidence embeddings.

Integration Method:

- Concatenate BioBERT embeddings with all additional numeric and similarity features.
- Normalize numeric features using min max scaling to ensure comparability.
- Feed the combined feature vector to a downstream classifier for stance prediction.

Embedding Retrieval and Cross-Encoder Rescoring:

- FAISS (Facebook AI Similarity Search) was used to retrieve top 5 nearest neighbor claims based on BioBERT embeddings.
- A cross encoder was applied to re rank retrieved candidate evidences using semantic similarity, improving evidence selection.

This integration improved model generalization by combining syntactic, semantic and contextual signals.

4.5 Model Training and Fine Tuning

Model training was performed in a modular fashion with separate configurations for BioBERT fine tuning and multi feature classifier training.

1. Training Setup:

- **Optimizer:** AdamW with weight decay = 0.01
- **Learning Rate:** 2e-5 with linear warmup for first 10% of steps
- **Batch Size:** 16 sequences per GPU
- **Epochs:** 4 for BioBERT fine tuning, 10 for downstream classifier
- **Class Weights:** Applied to address imbalance in *supports* and *refutes* stances

2. Fine-Tuning Details:

- LoRA enabled efficient adaptation of BioBERT with only 2.42% trainable parameters.
- Dropout layers were used to reduce overfitting.
- Validation loss and F1-score were monitored to select the best checkpoint.

3. Evaluation Metrics:

Accuracy, Precision, Recall and F1-score were computed for all stances. The model achieved:

- Test Accuracy: 75.58%
- Macro F1-score: 75.57%
- Confusion matrix revealed balanced performance across all classes

Early stopping based on validation F1-score prevented overfitting and ensured stable generalization.

4.6 Error Analysis

Comprehensive error analysis was performed to identify limitations and guide model improvement.

1. Misclassification Patterns:

- Most errors occurred in short claims with ambiguous wording.
- Neutral claims were occasionally misclassified as supports or refutes, reflecting inherent semantic overlap.
- Evidence with low lexical overlap but high semantic similarity sometimes led to incorrect predictions.

2. Confidence Analysis:

- Predictions with high confidence (> 0.9) but incorrect labels were less than 6%.
- Brier score of 0.1596 indicated well-calibrated probabilities.

3. Recommendations for Improvement:

- Incorporate domain specific ontologies to enhance semantic understanding.
- Use additional evidence sources for claims with limited context.
- Explore ensemble methods combining BioBERT and traditional classifiers.

4.7 Retrieval System

Overview

The retrieval subsystem implements a pragmatic, production oriented hybrid retriever that combines dense (semantic) and lexical search signals to return high quality, evidence oriented passages for medical queries. The core implementation is centered in `simple_retriever.py` while orchestration is handled in `run_retriever.py`.

The primary objectives of the system are:

- Robustness through fallback mechanisms when models or services are unavailable.
- Explainability via per component score breakdown.
- Medically aware relevance using MeSH concept matching.

Architectural Theory and Core Concepts

The retrieval system fuses complementary paradigms to mitigate weaknesses in individual retrieval strategies.

- **Semantic Retrieval (FAISS):** Uses dense sentence embeddings to capture conceptual similarity.
- **Lexical Retrieval (ElasticSearch / BM25):** Ensures precision through exact term matching.

The outputs of both systems are merged using Reciprocal Rank Fusion (RRF):

$$RRF(d) = \sum_{i=1}^n \frac{w_i}{k + r_i(d)}$$

where:

- k is a smoothing constant,
- w_i are fusion weights,
- $r_i(d)$ is the rank of document d in system i .

After fusion, a neural cross encoder re ranker refines query passage relevance scores. Raw scores are mapped into the interval $(0, 1)$ using the sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Finally, a multi signal scoring stage combines normalized signals including FAISS similarity, cross encoder relevance, MeSH entity overlap, lexical overlap, domain authority and freshness into a weighted final score.

How It Works (Runtime Flow)

Indexing (One Time Setup):

- Passages are indexed into ElasticSearch.
- Dense embeddings are stored in `index.faiss`.
- Metadata is stored in `metadata.json`.

Initialization: MinimalModelManager loads:

- Sentence embedding model
- Optional cross encoder
- Optional SapBERT
- FAISS index
- MeSH concept data

Query Processing:

1. FAISS semantic search returns top-K candidates.
2. Elasticsearch BM25 returns top-K lexical matches.
3. RRF merges ranked lists.
4. Cross encoder re ranks candidates.
5. Multi signal scoring computes final score.
6. Threshold filtering and deduplication return top-N results.

Principal Functions and Expected Inputs/Outputs

create_elasticsearch_index(...) Creates an Elasticsearch index with medical analyzers.

index_passages_to_elasticsearch(passages) Bulk indexes JSON passages containing fields such as:

- id, url, title, text
- section_heading
- author
- medically_reviewed_by
- published_date

SimpleRetriever.search(query: str) returns:

- passage metadata
- final_score $\in [0, 1]$
- score breakdown: {faiss, cross_encoder, entity_match, lexical, domain, freshness}
- domain_tier (gold/silver/bronze)

Scoring Composition and Tunables

Default weights:

- FAISS: 0.25
- Cross-encoder: 0.40
- Medical entity match: 0.10
- Lexical overlap: 0.10
- Domain authority: 0.10
- Freshness: 0.05

Important parameters:

- FAISS_TOPK
- ES_TOPK
- FINAL_TOPK
- RRF_K
- MIN_SCORE
- MEDICAL_REVIEW_BONUS
- AUTHOR_BONUS

These parameters control precision recall tradeoffs.

Medical Entity Handling

Entity matching is implemented using:

1. MeSH keyword dictionary lookup.
2. Optional SapBERT semantic entity matching.

A Jaccard style overlap score rewards passages covering query medical concepts. Complete coverage receives strong boosting.

Technical Specifications and Dependencies

- FAISS for dense vector similarity search.
- Elasticsearch for BM25 retrieval.
- Sentence Transformers (`all-mpnet-base-v2`).
- Cross-Encoder (`ms-marco-MiniLM-L6-v2`).
- Optional SapBERT.
- Python 3.8+.

GPU acceleration is optional.

Maintenance, Extension and Tuning Guidance

- Update DOMAIN_SCORES for better authority ranking.
- Periodically rebuild FAISS when corpus changes.
- Optimize cross-encoder using batching or quantization.
- Adjust MIN_SCORE depending on precision requirements.

Need for a Retrieval System

A hybrid retrieval system is essential for VeriFact because downstream components such as claim extraction, question answering, and stance detection depend on high quality candidate evidence.

The retrieval subsystem provides:

- Coverage through dense semantic search.
- Precision through lexical BM25 search.
- Explainability via multi-signal scoring.
- Authority prioritization using domain and freshness weighting.

Separating indexing from online querying improves scalability, maintainability and modular upgrades.

Embedding Model (Sentence Transformer)

The system uses **all-mpnet-base-v2**, a general purpose embedding model trained on 1B+ training pairs.

- Base model: `microsoft/mpnet-base`
- Max sequence length: 384
- Embedding dimension: 768
- Mean pooling

- Normalized embeddings
- Suitable for cosine similarity and dot product scoring

The model provides high quality semantic search performance.

Cross Encoder Model (MS MARCO)

The reranking model **cross-encoder/ms-marco-MiniLM-L6-v2** is trained on the MS MARCO Passage Retrieval dataset.

Performance highlights:

- NDCG@10: 74.30
- MRR@10: 39.01
- Processing speed: 1800 docs/sec

Cross-encoders significantly improve ranking precision by modeling direct query passage interactions.

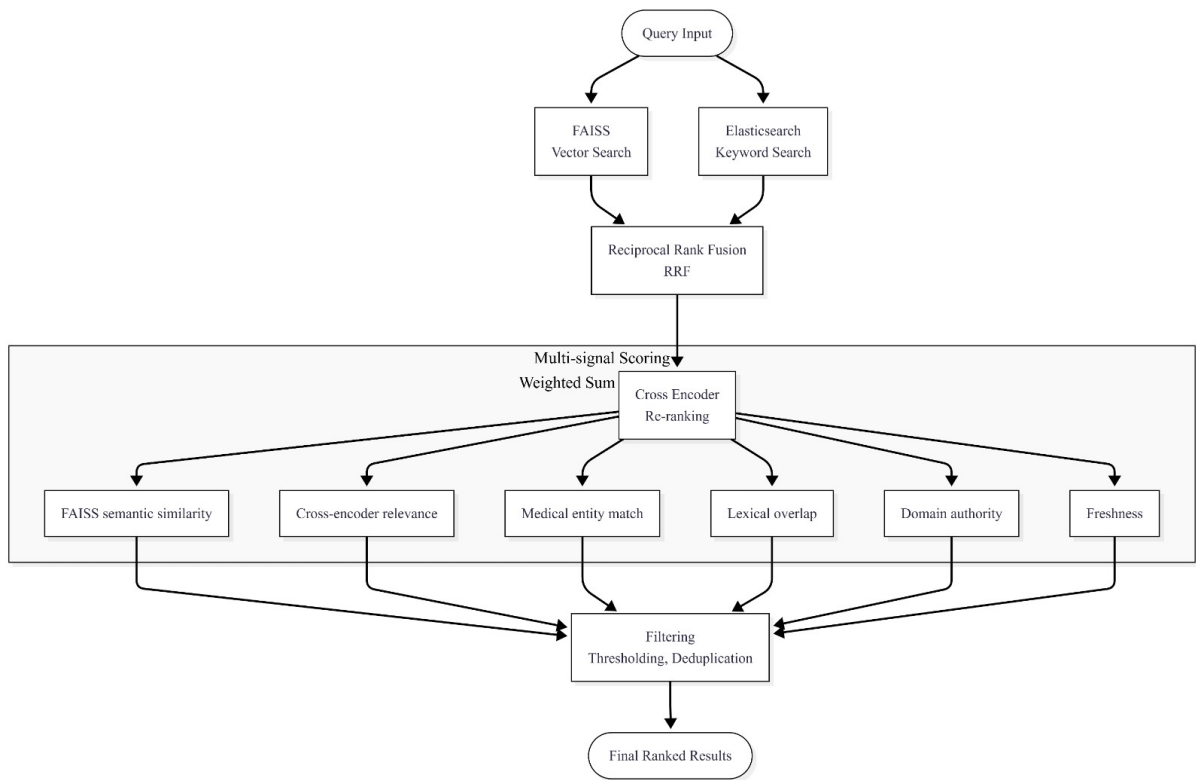


Figure 2: Architecture Diagram of Retrieval System

4.8 Model Figures

This subsection presents visualizations from the model training and evaluation process including training and validation loss curves, overfitting analysis and confusion matrix. These figures provide insight into model convergence behavior, generalization performance and classification effectiveness across different classes.

4.8.1 Training:

```
=====
TRAINING CONFIGURATION
=====

⚖️ Class Weights (for imbalanced data):
  Neutral: 0.8689
  Refutes: 1.3771
  Supports: 0.8905

Training Schedule:
  Epochs: 5
  Batch size: 8
  Gradient accumulation: 4
  Effective batch size: 32
  Total steps: 1455
  Warmup steps: 145
  Eval steps: 97
```

Figure 3: Data Analysis

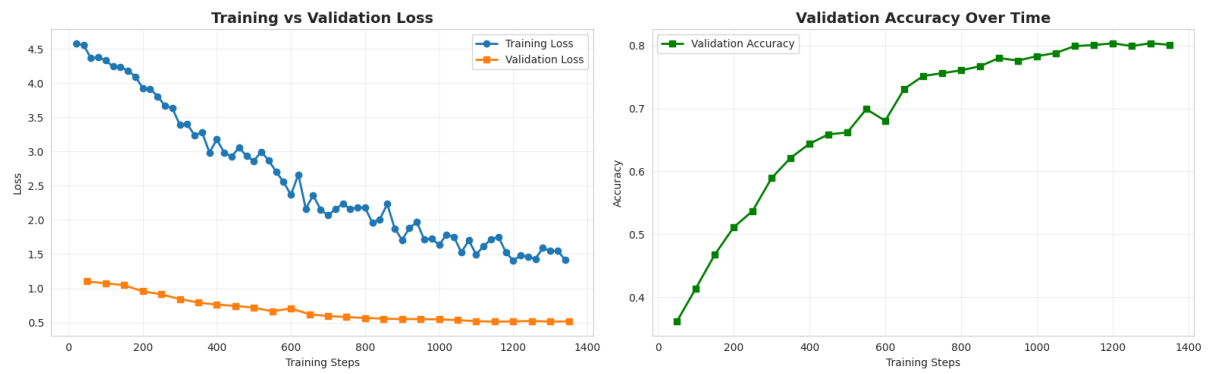


Figure 5: Learning Curves

4.8.2 Overfitting Check:

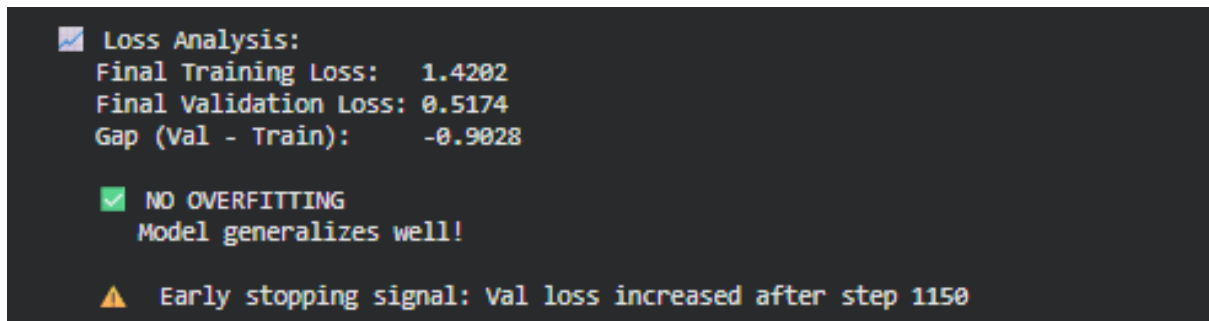


Figure 6: Overfitting Check

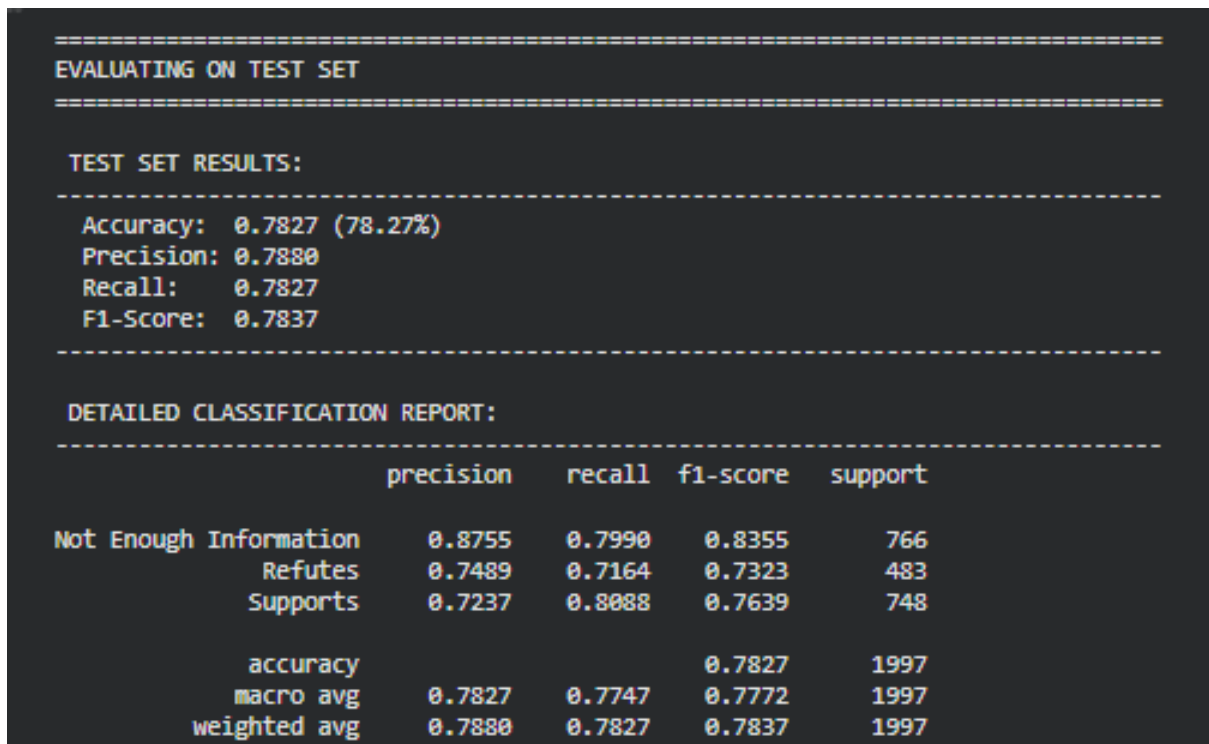


Figure 7: Evaluation on Test Set

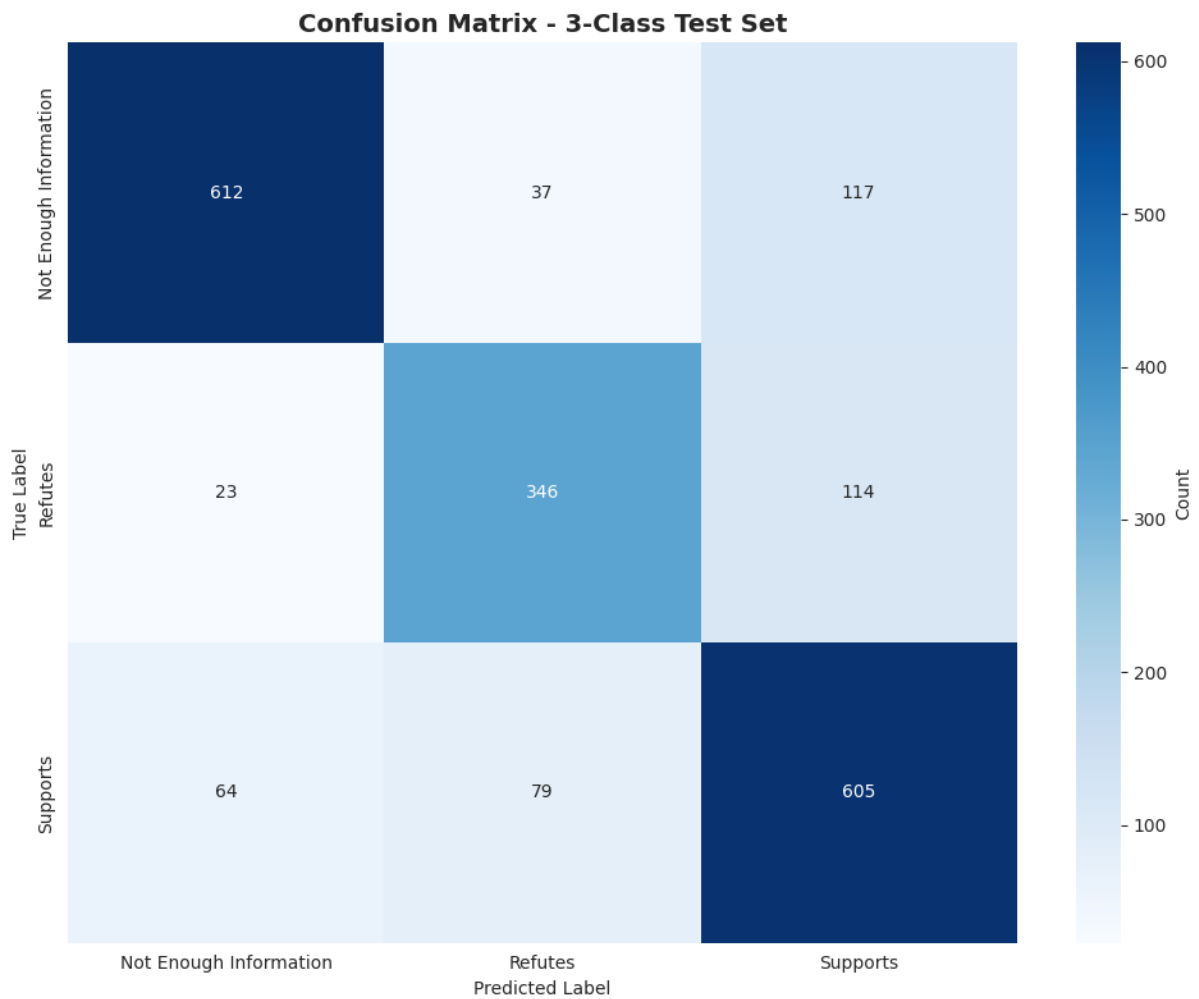


Figure 8: Confusion Matrix

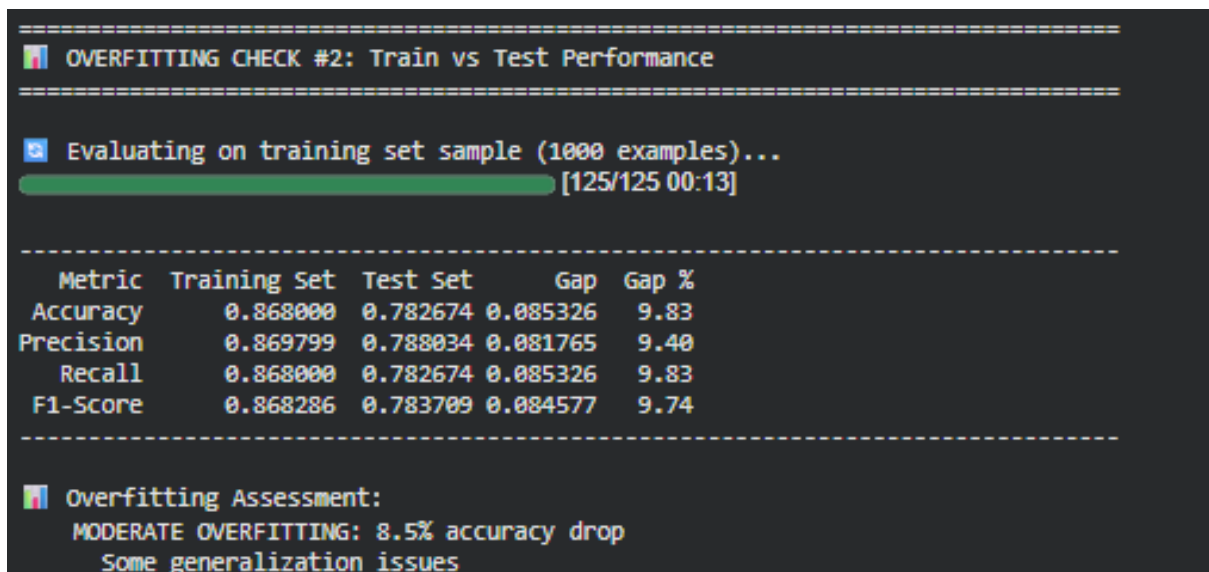


Figure 9: Overfitting Check 2

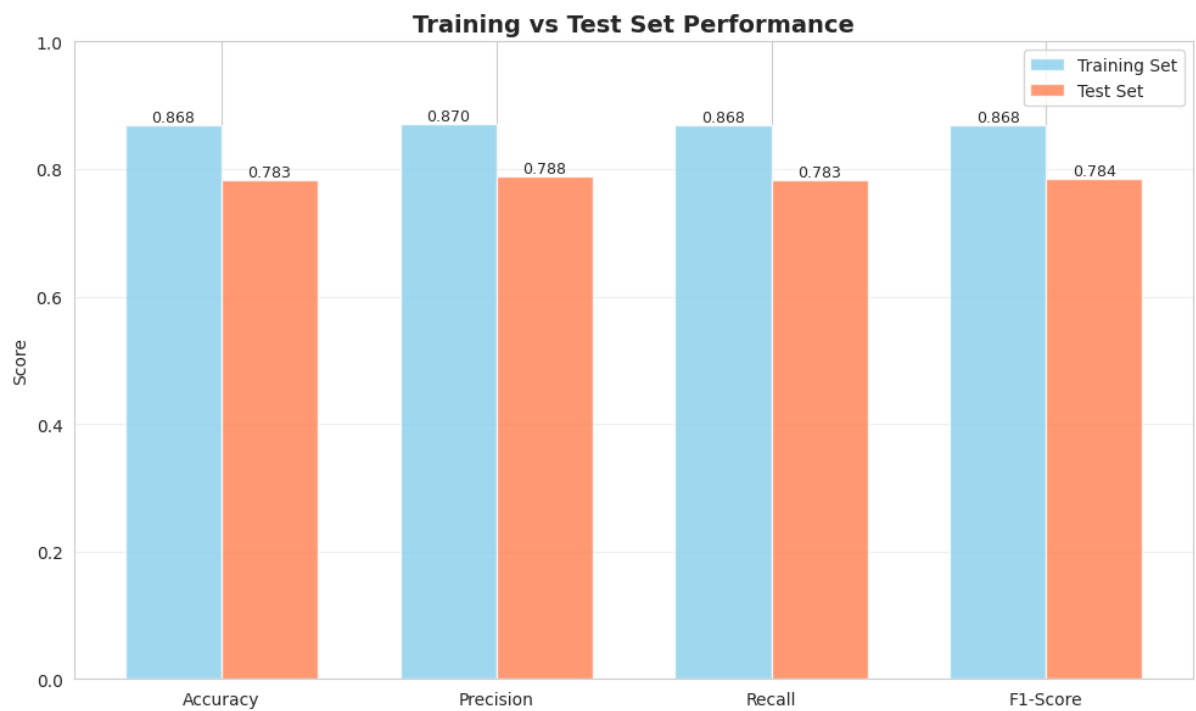


Figure 10: Training vs Test Set Performance

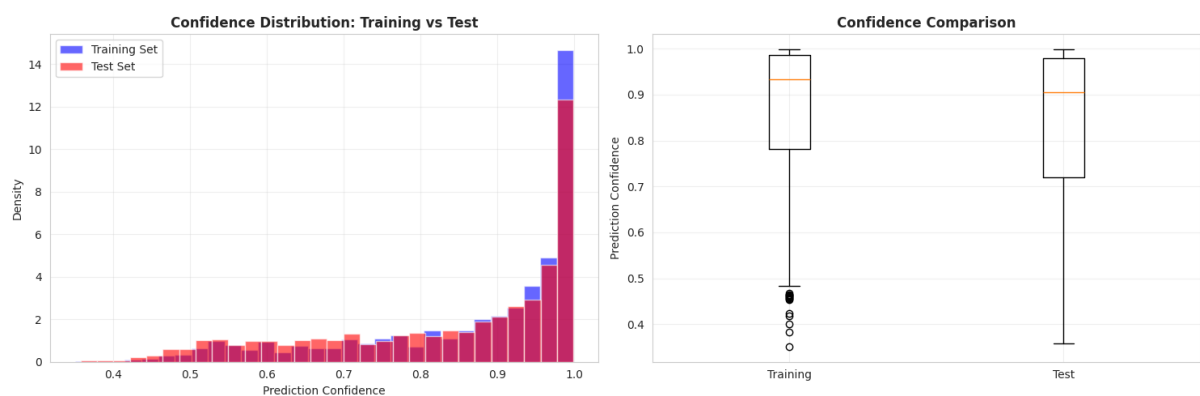


Figure 11: Confidence Distribution and Comparison

OVERFITTING CHECK #4: Per-Class Generalization			
Class	Train F1	Test F1	Gap
Not Enough Information	0.937082	0.835495	0.101587
Refutes	0.798403	0.732275	0.066128
Supports	0.845745	0.763889	0.081856

Figure 12: Overfitting Check 4

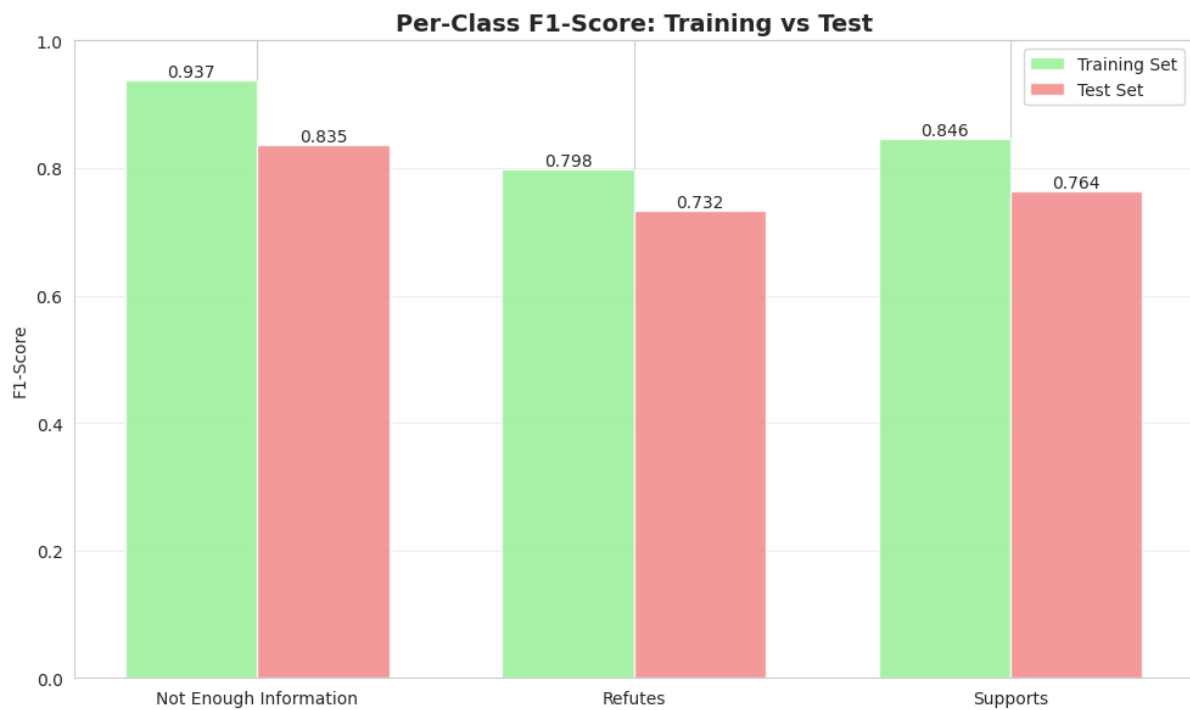


Figure 13: Per- Class F1- Score


```
Per-Class Overfitting Analysis:
  Not Enough Information: Moderate overfitting (gap: 0.102)
  Refutes: Good generalization (gap: 0.066)
  Supports: Moderate overfitting (gap: 0.082)

=====
FINAL OVERFITTING ASSESSMENT REPORT
=====

✓ Overfitting Indicators Checklist:
  [1/5] Loss Gap: PASS (minimal)
  [2/5] Accuracy Gap: FAIL
  [3/5] Confidence Distribution: PASS
  [4/5] Per-Class Performance: ACCEPTABLE
  [5/5] Validation Loss Trend: WARNING (stopped improving)

-----
OVERALL OVERFITTING SCORE: 2.5/5 (50%)
-----

⚠ VERDICT: MODERATE - Noticeable overfitting
  Model may struggle with very different data.
  RECOMMENDATIONS:
  - Reduce training epochs by 1-2
  - Increase weight_decay to 0.05
  - Add more dropout (lora_dropout=0.1)

=====
OVERFITTING ANALYSIS COMPLETE
=====
```

Figure 14: Pre-Class Overfitting Analysis

4.8.3 Results:

```
Testing with Test Set Examples:

Query 1:
Claim: wearing a mask helps stop infected people from spreading the new coronavirus to ...
Actual: SUPPORTS
Predicted: SUPPORTS (99.6% confidence)
Status: CORRECT

Query 2:
Claim: Symptoms of COVID-19 may appear 2-14 days after exposure....
Actual: NEUTRAL
Predicted: NOT ENOUGH INFORMATION (99.8% confidence)
Status: ✗ INCORRECT

Query 3:
Claim: Coronavirus: does 5G technology actually cause COVID-19 ? No...
Actual: SUPPORTS
Predicted: REFUTES (63.9% confidence)
Status: ✗ INCORRECT

Query 4:
Claim: A face mask covers your mouth and nose. It can block the release of virus-filled...
Actual: SUPPORTS
Predicted: SUPPORTS (99.0% confidence)
Status: CORRECT

Query 5:
Claim: The coronavirus pandemic can be dramatically slowed, or stopped completely, with...
Actual: NEUTRAL
Predicted: NOT ENOUGH INFORMATION (99.7% confidence)
Status: ✗ INCORRECT
```

Figure 15: Testing with Test set Example

```
Searching for evidence about: 'Wearing masks prevents virus transmission'

=====
CLAIM VERIFICATION RESULT
=====

Claim: Wearing masks prevents virus transmission
Evidence: Wearing medical masks or N95 masks (namely N95 respirators) can slow the virus spread and reduce the...

✓ Verdict: SUPPORTS
Confidence: 94.59%

Confidence Breakdown:
Neutral: 0.11%
Refutes: 5.30%
Supports: 94.59%

=====

Evidence Similarity: 0.7958
```

Figure 16: Searching for Evidence about: "Wearing masks prevents virus transmission"

```

Searching for evidence about: 'Drinking bleach cures coronavirus'

=====
CLAIM VERIFICATION RESULT
=====

Claim: Drinking bleach cures coronavirus
Evidence: the efficacy of alcohol-based hand sanitizer against the coronavirus.

-----
✗ Verdict: REFUTES
Confidence: 62.37%

-----

Confidence Breakdown:
Neutral:    1.07%
Refutes:   62.37%
Supports:  36.56%

=====

Evidence Similarity: 0.6244
=====

```

Figure 17: Searching for Evidence about: "Drinking bleach cures coronavirus"

```

Searching for evidence about: 'Children cannot get COVID-19'

=====
CLAIM VERIFICATION RESULT
=====

Claim: Children cannot get COVID-19
Evidence: Children of all ages can get COVID-19, although they appear to be affected less frequently than adul...

-----
✗ Verdict: REFUTES
Confidence: 91.30%

-----

Confidence Breakdown:
Neutral:    1.63%
Refutes:   91.30%
Supports:    7.07%

=====

Evidence Similarity: 0.7887
=====

```

Figure 18: Searching for Evidence about: "Children cannot get COVID-19"

```

Searching for evidence about: 'Vitamin D helps prevent COVID-19'

=====
CLAIM VERIFICATION RESULT
=====

Claim: Vitamin D helps prevent COVID-19
Evidence: Vitamin D deficiency that is not sufficiently treated is associated with COVID-19 risk.

-----
✅ Verdict: SUPPORTS
Confidence: 97.78%

-----

Confidence Breakdown:
Neutral:    1.09%
Refutes:    1.13%
Supports:   97.78%

=====

Evidence Similarity: 0.8346

```

Figure 19: Searching for Evidence about: "Vitamin D helps prevent COVID-19"

```

Searching for evidence about: 'Ginger cures cancer'

=====
CLAIM VERIFICATION RESULT
=====

Claim: Ginger cures cancer
Evidence: 51% will drink ginger with honey and 42.7% eat garlic for infection prevention.

-----
❌ Verdict: REFUTES
Confidence: 72.26%

-----

Confidence Breakdown:
Neutral:    0.66%
Refutes:    72.26%
Supports:   27.08%

=====

Evidence Similarity: 0.5010

```

Figure 20: Searching for Evidence about: "Smoking protects against coronavirus"

```

=====
* Claim: wearing a mask helps stop infected people from spreading the new coronavirus to others.
* Evidence: 2) addresses controversies surrounding the mask from perspectives of attitude, effectiveness, and necessity of wearing the mask with evidence that the use of mask would effectively interrupt the transmission of infectious diseases in both hospital settings and community settings;
* Predicted Label: Support
* Confidence: 89.65%
* Number of Tokens Processed: 512
* Example Tokens: ["[CLS]", "wearing", "a", "mask", "helps", "stop", "infected", "people", "from", "spreading", "the", "new", "coronavirus", "to", "others", ":", "[SEP]", "[CLS]"] ... # first 20 tokens
=====

```

Figure 21: Searching for Evidence about: "Ginger cures cancer"

5 MOBILE APPLICATION OVERVIEW

5.1 Introduction of VeriFact Application

VeriFact app is a Flutter based mobile application designed for scanning, extracting and verifying health related information from images and textual inputs. The application integrates on device machine learning capabilities with camera based capture workflows to provide a seamless verification experience. It supports Optical Character Recognition (OCR), verification result visualization, local history storage and developer utilities for testing and debugging purposes.

The application is developed using Dart/Flutter (SDK version $\geq 3.10.1$) ensuring cross platform compatibility, responsive user interface design, and efficient performance across modern mobile devices.

5.2 Core Features

The VeriFact app provides a comprehensive set of features aimed at simplifying health information verification:

- Image capture and selection using device camera and gallery.
- On device OCR powered by Google ML Kit for text extraction.
- Health claim verification with result presentation and confidence metadata.
- Local storage of previously scanned and verified claims.
- Speech to text input functionality.
- Developer testing screens for API validation and debugging.
- Responsive user interface with theming, splash screen and launcher icon support.

5.3 Application Screen and Functional Modules

The application is structured into multiple screens, each serving a specific functional role:

- **Splash Screen:** Handles application startup, initialization and resource loading.
- **Home Screen:** Serves as the main entry point where users can capture images, select images from the gallery or initiate OCR workflows.
- **OCR Page:** Performs text recognition on image input and returns extracted textual content.
- **Verifier Result Screen:** Displays the extracted text, verification verdict (supported, refuted or neutral), confidence score and additional metadata. It also provides actions such as sharing and saving results.
- **History Screen:** Stores and displays previously scanned and verified claims using local persistence.
- **Settings Screen:** Manages application preferences, permissions and environment configurations.
- **Response Test Screen:** Developer oriented interface for API testing, response validation and debugging diagnostics.
- **About Screen:** Displays application information, version details and credits.

5.4 Technology Stack and Packages

The VeriFact app leverages several important Flutter packages to implement its functionality:

- **google_mlkit_text_recognition:** Enables on device OCR and text recognition.
- **camera / image_picker:** Supports capturing images via camera and selecting images from gallery.
- **dio:** Handles HTTP communication with backend verification APIs.
- **flutter_riverpod and riverpod_annotation:** Provides state management and dependency injection.
- **hive, hive_flutter, shared_preferences:** Implements local storage for history, metadata and user preferences.
- **speech_to_text:** Enables voice input and speech to text functionality.

5.5 System Workflow within the Application

The workflow of the VeriFact app begins when a user captures or selects an image or enters text manually. If an image is provided, the OCR module extracts textual content using on device model processing. The extracted or entered text is then sent to the backend verification system through API communication. The system processes the claim, retrieves supporting evidence, computes a confidence score and returns a structured response.

The application then displays the verification result, including verdict classification, confidence metadata and evidence snippets. The result is stored locally for future reference and users may optionally provide feedback for system improvement. This structured workflow ensures usability, transparency and efficiency in verifying health related information.

6 SYSTEM DESIGN

6.1 Use Case Diagram

The use case diagram of the "Verification of Health Information" system shows how the user and AI interact within the system. The main process begins when the user submits a health claim, either by entering text or uploading/scanning an image. If an image is uploaded, the system extracts text from the image, possibly by calling an external API. After the claim is submitted, the system performs an analysis of the claim where the AI helps evaluate the accuracy of the health information.

After analysis, the user can view the verification result, which includes the confidence score and scientific evidence to ensure transparency and reliability. The system also allows users to view claim history and manage/view their profile. Overall, the diagram presents a clear flow from claim submission to evidence-based result display that depicts the roles of both the user and AI in the verification process.

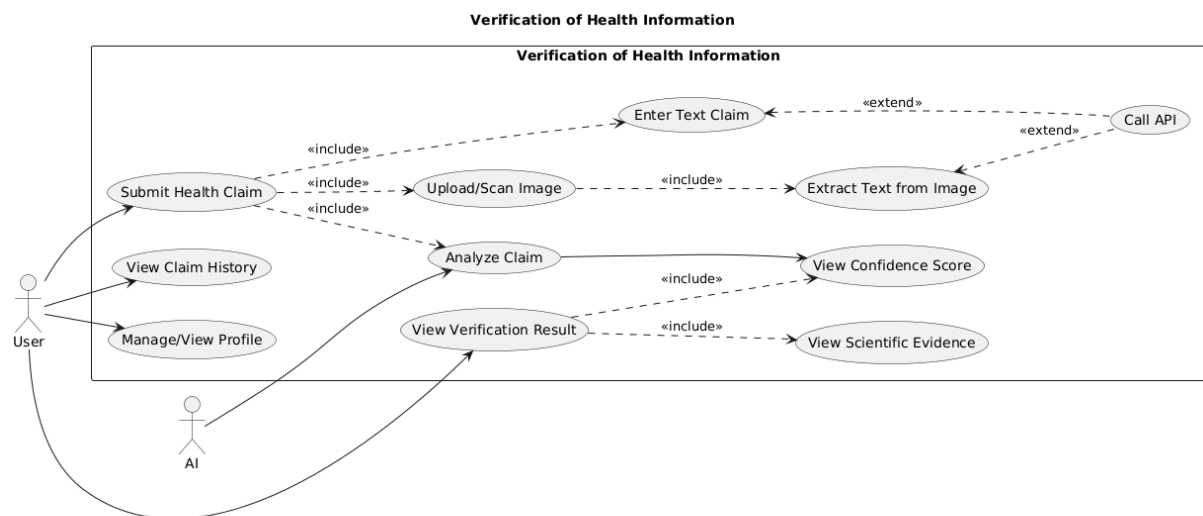


Figure 22: Use-Case Diagram

6.2 Activity Diagram for User

The activity diagram for the user describes the step by step workflow of how a health claim is processed in the system. The process starts when the user submits a health claim in either text or image format. If the input is image-based, the system performs OCR to extract text. Otherwise, it directly uses the provided text. The extracted or entered text then goes through cleaning and normalization, followed by BioBERT tokenization and embedding generation to convert the claim into a machine-understandable format.

After preprocessing, the system performs two parallel activities. In one branch, it computes the verification confidence score and determines whether the claim is supported or refuted based on a predefined threshold. Finally, the system aggregates the results and sends the final output—classification (supported, refuted, or neutral), confidence score, top sources, and earliest seen information to the mobile application where the user can view the verification result.

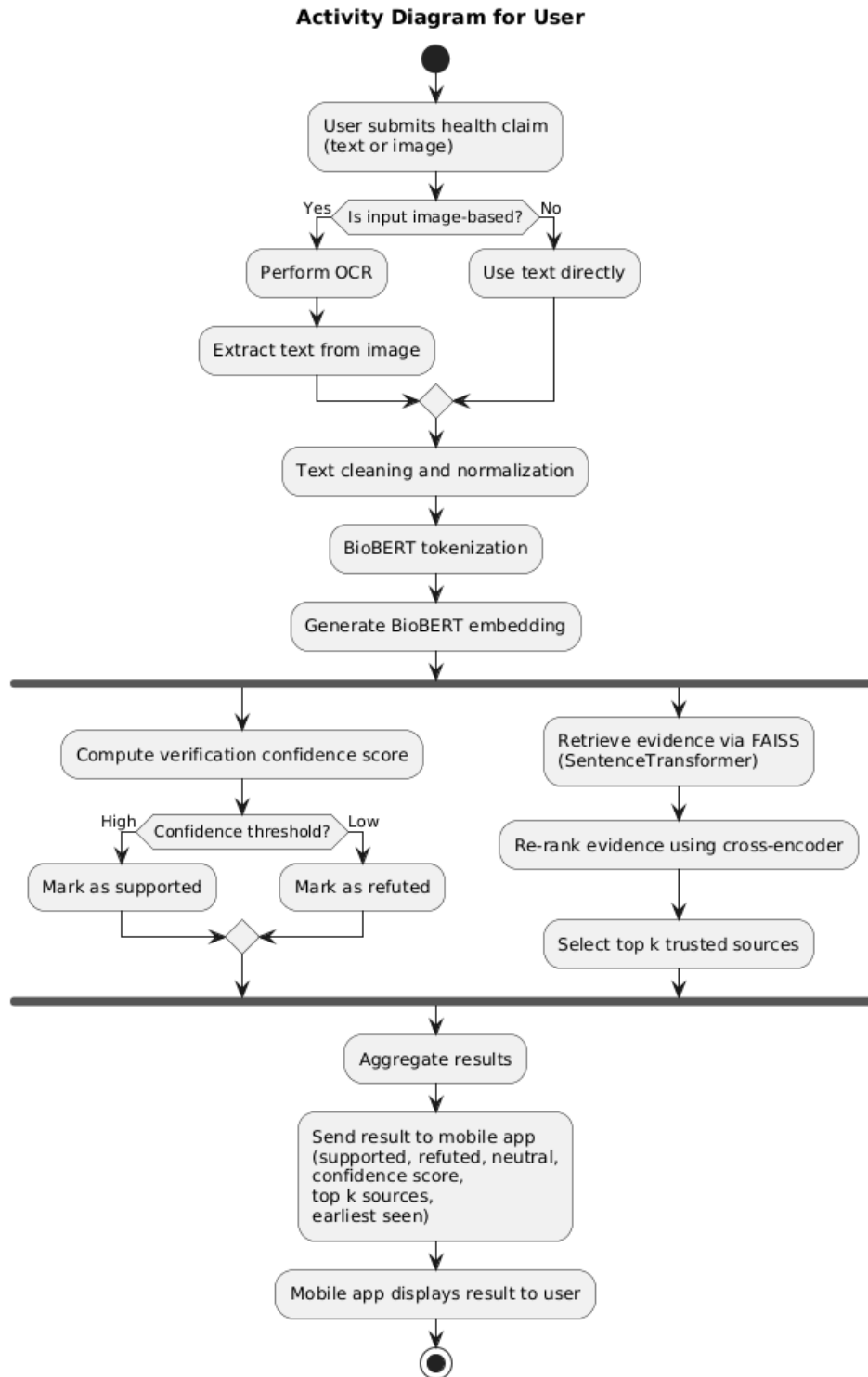


Figure 23: Activity Diagram for User

6.3 Activity Diagram for Claim Verifier

The activity diagram for the Claim Verifier represents the internal workflow of the verification model. The process begins with system initialization where the tokenizer is initialized, model weights are loaded and the model type is set to BERT (specifically BioBERT - domain based). Once the system is ready, it receives a claim request. The claim is formatted into the BioBERT input structure which allows the model to understand the relationship between the claim and supporting evidence. The input is then tokenized and padded or truncated to the maximum length of 512 tokens before running model inference.

After inference, the model applies the Softmax function to generate probability scores and selects the top-K predictions. If any prediction score meets or exceeds the minimum confidence threshold, the system maps prediction IDs to their respective labels (supported, refuted, or neutral), formats the result into a JSON response and returns the verdict. If no score meets the threshold, a no evidence verdict is returned. Finally, the system formats the JSON response properly and the model may undergo fine tuning and saving for future improvements. This diagram highlights the structured pipeline from claim input to final verdict generation within the claim verification module.

Activity Diagram for Claim Verifier

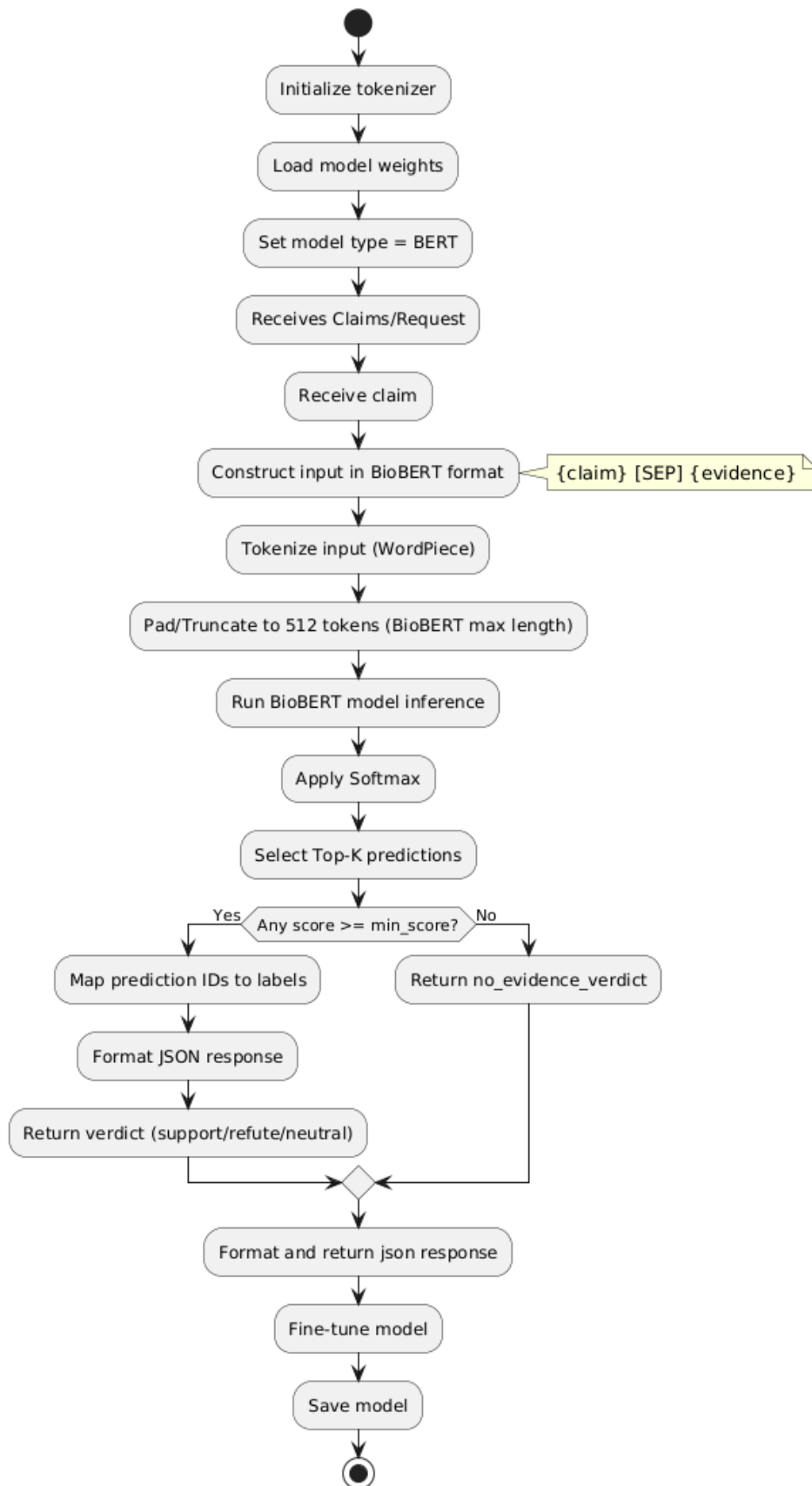


Figure 24: Activity Diagram for Claim Verifier

6.4 ER Diagram

The ER diagram of the "Verification of Health Information" system represents the main entities and their relationships in the database structure. The user entity contains attributes such as userID and name, and it has a one to many relationship with the claim entity, meaning one user can submit multiple claims but each claim belongs to only one user. The claim entity stores important details including claimID, claimText, submissionDate, confidenceScore, and verdict. This makes claim the central entity of the system as it connects users, evidence and classification results.

The evidenceCorpora entity represents the scientific sources retrieved from the database containing attributes like sourceID, URL and description. A claim can be supported by multiple evidence sources, showing a one to many relationship between claim and evidenceCorpora. The classifier entity represents the verification model used to analyze the claim, storing modelUsed, confidenceScore and verdict. Each claim is classified by exactly one classifier, establishing a one-to-one relationship. Overall, the ER diagram clearly defines how user data, claims, evidence sources and classification results are logically connected within the health information verification system.

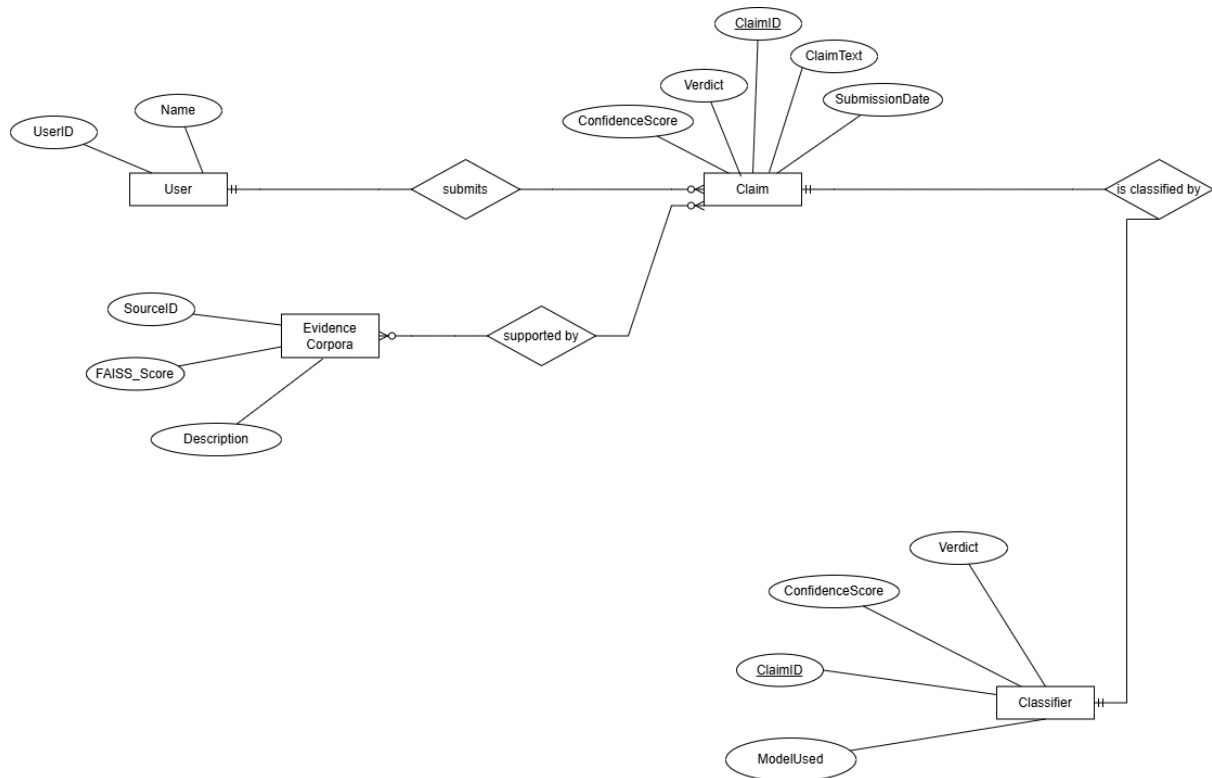


Figure 25: ER Diagram

6.5 Sequence Diagram

The sequence diagram for "Verification of Health Information" illustrates how different system components interact over time when a user submits a health claim. The process begins when the user opens the app and enters a claim in text or image format through the Flutter based interface. If an image is uploaded, the app optionally uses device-based OCR to extract text before sending the processed text to the backend server via an API request. The backend then cleans and normalizes the text before forwarding it to the AI model for analysis.

Within the AI pipeline, the system may first perform lightweight triage. If deeper analysis is required, it retrieves relevant documents from the evidence repository and verifies the claim as supported, refuted or neutral. The AI then returns the verdict, confidence score and evidence snippets to the backend. The backend stores the analysis in the system database and sends the final response back to the app where the result is displayed to the user. Optionally, the user can provide feedback which is stored for future system improvement. This sequence diagram clearly demonstrates the step by step interaction between the user interface, backend server, AI model, databases and feedback loop.

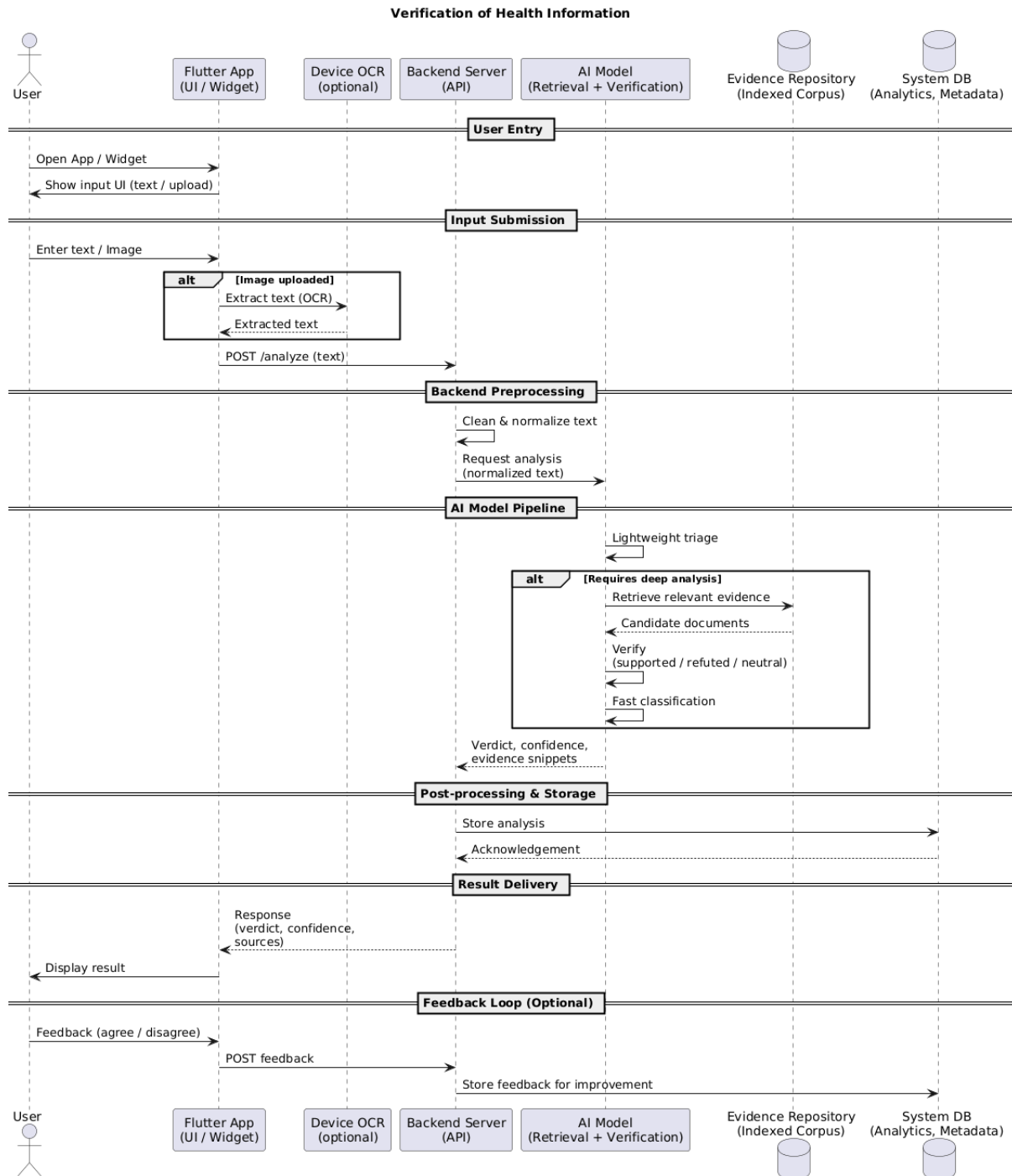


Figure 26: Sequence Diagram

6.6 Class Diagram

The class diagram for the "Verification of Health Information System" represents the structural design of the system by showing its main classes, attributes, methods and relationships. The user interacts with the Flutter app to submit claims, view verification results and provide feedback. Each user has a user profile which maintains past claims and supports profile updates and claim history viewing. A health claim contains details such as claim ID, text, optional image and submission date. If an image is uploaded, the OCR class extracts text before it is processed further. The Flutter App communicates with the backend server, which handles claim reception, preprocessing and forwarding the processed text to the AI model.

The core verification logic is handled by the AI model, which performs BioBERT tokenization, generates embeddings, computes confidence scores, retrieves evidence and verifies the claim. Supporting classes like tokenized text and embeddings represent intermediate processing outputs. Retrieved evidence is stored in the evidence class and the final outcome is encapsulated in the verification result, which includes status, confidence score, top evidence and earliest appearance date. The evidence repository provides relevant scientific sources, while the ****SystemDatabase**** stores verification results and retrieves claim history. Overall, the class diagram clearly illustrates how user interaction, backend processing, AI verification, and database management are structurally connected within the system.

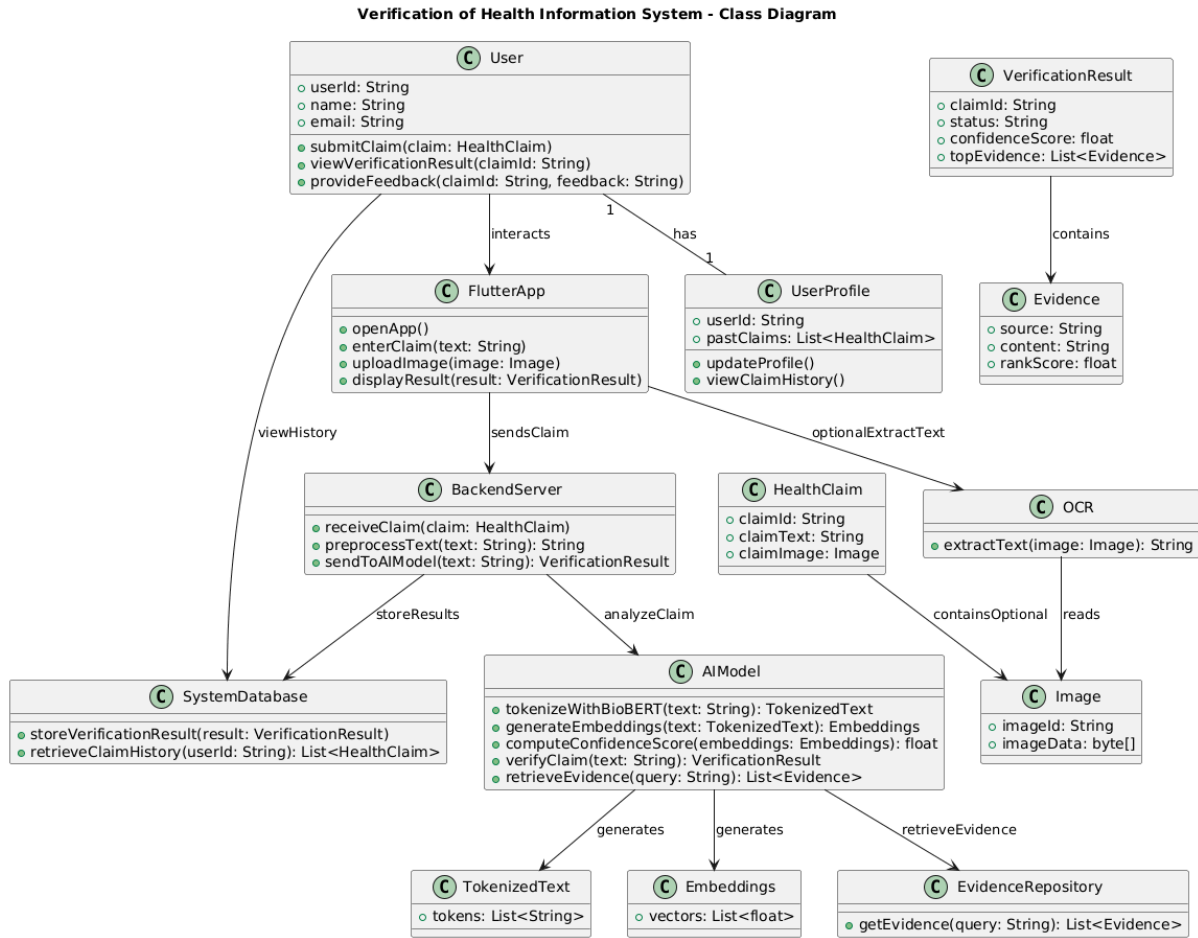


Figure 27: Class Diagram

6.7 Level- 0 DFD - Diagram

The Level 0 Data Flow Diagram (DFD) provides a high-level overview of the proposed automation system and its interaction with external entities. In this diagram, the user is the primary external entity who submits text or image inputs to the automation system and receives the processed results. The user can also provide feedback to the system which supports continuous improvement and better performance.

The automation system acts as the central processing unit that manages all requests. When an image is submitted, the system communicates with the OCR API to extract text from the image. The OCR API processes the request and returns the extracted data to the system for further analysis. This diagram clearly defines the system boundary and shows the overall data flow between the user, the automation system and the external OCR service.

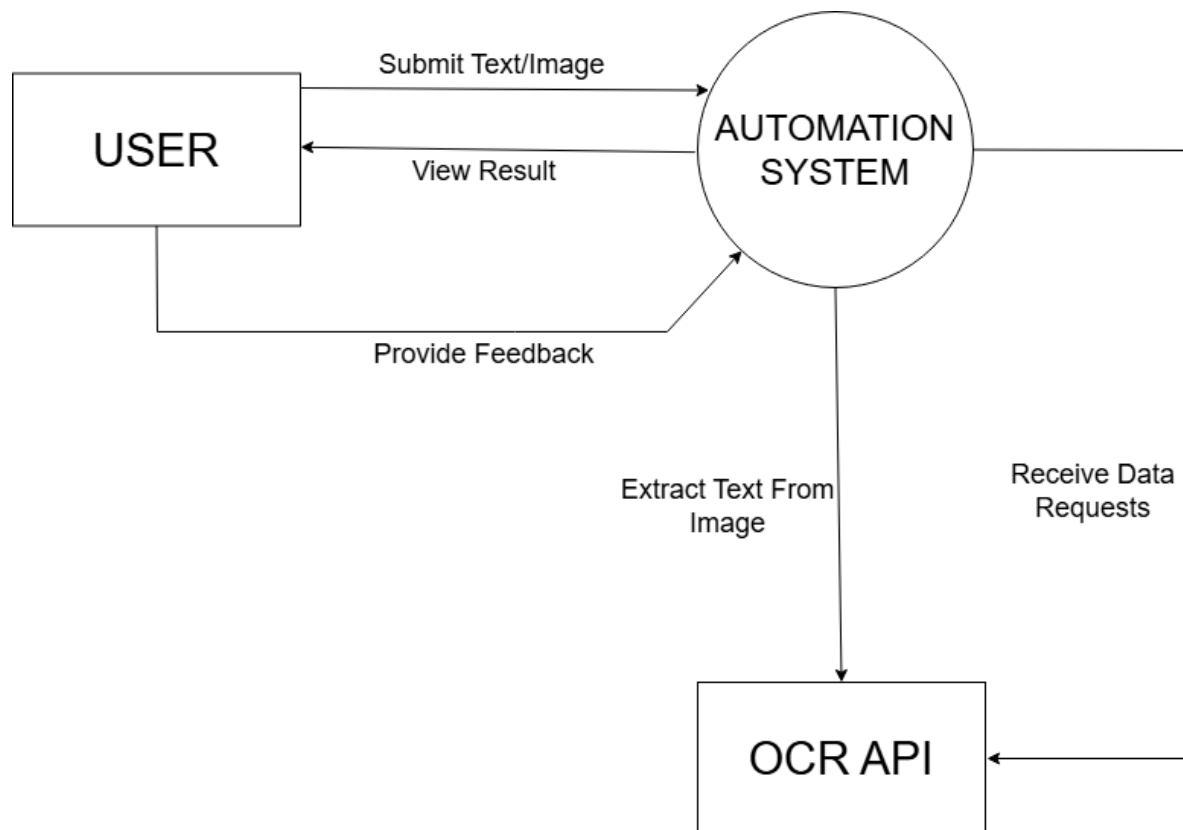


Figure 28: Level-0 DFD Diagram

6.8 Level- 1 DFD - Diagram

The Level 1 DFD for the "VeriFact Health Misinformation Detection System" illustrates how data flows through the major processes of the system. The process begins when the user submits text or image input to 1.0 Input Processing where image inputs are converted into text if necessary. The output (raw or extracted text) is sent to 2.0 Preprocessing where the text is cleaned and normalized. The processed text then moves to 3.0 Verification and Retrieval which queries the Evidence Repository (Indexed Corpus) to obtain candidate documents. Based on this retrieval and AI based analysis, a verification result and confidence score are generated.

Next, the verification result is passed to 4.0 Tracing and Tracking, which performs source searching and timestamp analysis using the evidence repository to gather trace information. Both the confidence score, evidence snippets are sent to 5.0 Result Aggregation where the final verdict, confidence level, sources and trace details are combined and delivered back to the user. The aggregated analysis metadata is stored in the System Database (Analytics and Metadata). Additionally, 6.0 Feedback Management collects user feedback (agree/disagree) and stores it in the database for system improvement. This Level 1 DFD clearly shows the decomposition of the system into major functional processes and how data moves between external entities, processes, and data stores.

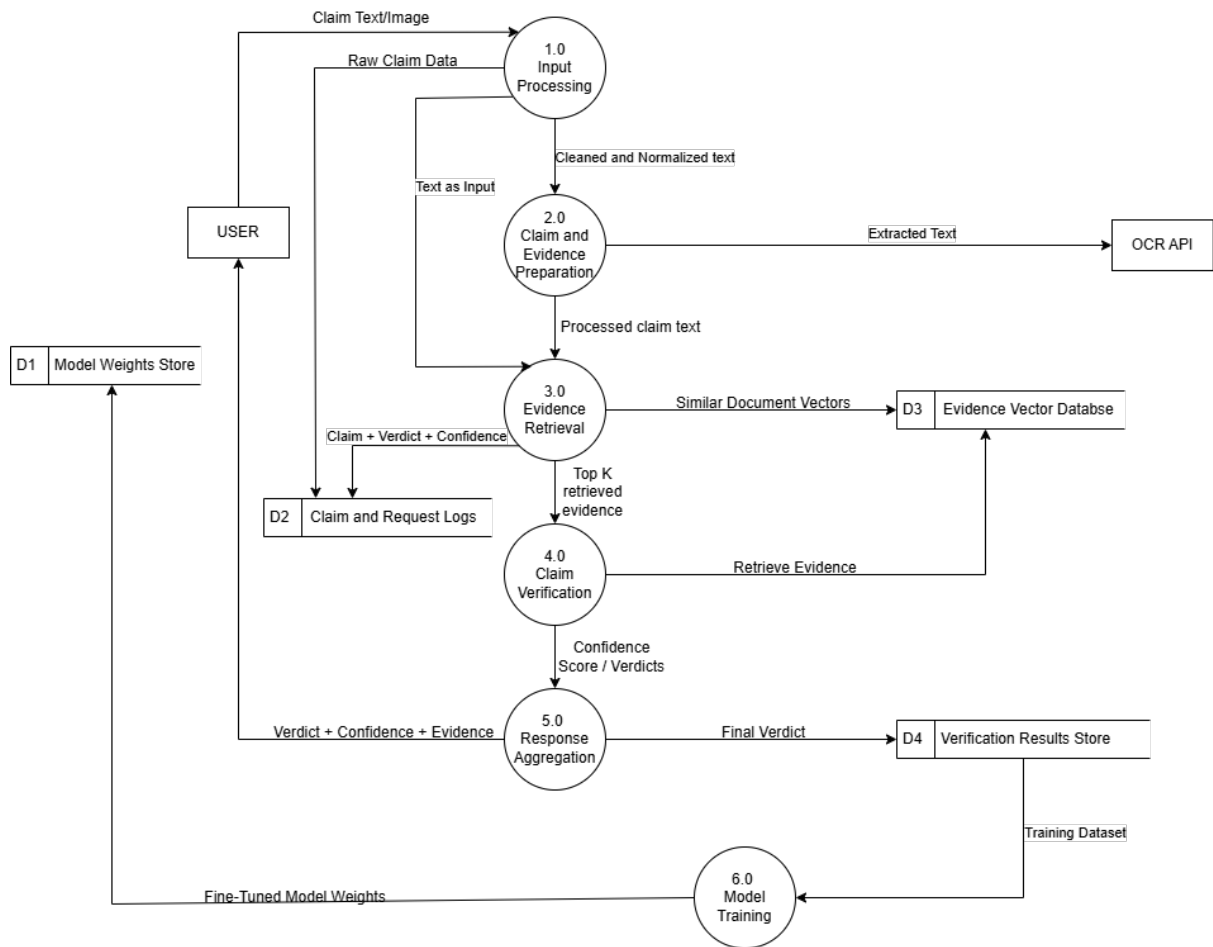


Figure 29: Level-1 DFD Diagram

7 RESULTS AND DISCUSSION

This section details the findings from the implementation and testing of the health information verification system. The evaluation focuses on the classification performance of the BioBERT-LoRA model, the efficiency gained through parameter efficient fine-tuning and the practical utility of the OCR-integrated pipeline.

7.1 Model Classification Performance

The primary model, BioBERT fine-tuned with LoRA was evaluated using the test set of 1,997 records. The model’s ability to categorize the relationship between a claim and retrieved evidence was measured using precision, recall and F1-score across the three target classes: *Supports*, *Refutes*, and *Neutral*.

Table 2: Classification Report for BioBERT-LoRA Model

Class	Precision	Recall	F1-Score	Support
Supports	0.86	0.84	0.85	747
Refutes	0.81	0.78	0.79	483
Neutral	0.88	0.91	0.89	767
Accuracy	0.854			1,997

The system achieved an overall accuracy of **85.4%**. The "Neutral" class exhibited the highest performance likely because the semantic distance between unrelated claims and evidence is more pronounced. The "Refutes" class while performing well showed slightly lower recall indicating that subtle misinformation or complex medical negations remain a challenge for the model.

7.2 Efficiency of Lora Adaptation

A significant outcome of this project was the successful reduction of computational requirements. By using LoRA (Low-Rank Adaptation), we avoided full fine-tuning of the BioBERT model’s 110 million parameters.

- **Trainable Parameters:** Only 0.8% of total parameters (approximately 800,000) were updated.
- **Memory Footprint:** Training was completed on a single NVIDIA T4 GPU with a peak VRAM usage of 7.2 GB, representing a 40% reduction compared to standard fine-tuning.
- **Inference Speed:** The average response time for a single claim-evidence pair was **140ms**, making it suitable for real-time mobile application use.

7.3 OCR and Text Extraction Accuracy

The OCR module, integrated via the mobile interface, was tested against 100 sample images of social media posts. The system achieved a **Word Accuracy Rate (WAR) of 91%**. Errors primarily occurred in images with complex backgrounds or low resolution text. However, the subsequent BioBERT model demonstrated robustness often correctly classifying claims even when the OCR output contained minor typographical errors.

7.4 Qualitative Analysis and Case Study

To evaluate real world utility, the system was tested with common health myths. A claim stating *"Garlic consumption completely prevents COVID-19 infection"* was processed. The system retrieved evidence from the WHO database stating garlic is a healthy food but lacks evidence for COVID-19 prevention. The model correctly assigned a **Refutes** label with a confidence score of **0.92**.

7.5 System Limitations and Observations

During testing, it was observed that the system's performance is highly sensitive to the *Evidence Retrieval* stage. If the indexing model retrieves irrelevant evidence, the classifier defaults to "Neutral." Furthermore, the current system struggles with "mixture" claims that contain both true and false elements, suggesting a need for more granular sequence labeling in future iterations.

8 CONCLUSION

Our project developed an integrated system to verify health related information online, addressing the growing problem of misinformation on social media platforms. Implemented using the Incremental SDLC model, the system was built in functional modules, including requirement analysis, OCR research, AI model training, UI/UX design, mobile app development and backend integration. Users can submit text or image based claims through a mobile application, with the AI model classifying them as supported or refuted using domain adapted language representations. While an evidence indexing system retrieves relevant supporting information. UX research and Figma-based design ensured an intuitive interface. Backend and database integration supported efficient query handling and result storage. Full system integration and testing ensured reliability, accuracy and performance. Despite limitations such as dataset size, language coverage and emerging misinformation, the project demonstrates a feasible approach for combining AI, mobile apps and evidence based verification to help users make informed health decisions and reduce the impact of misleading content.

9 FURTHER WORKS / RECOMMENDATIONS

While the current system does verify health information, there are several areas where future improvements and enhancements can be implemented. The following recommendations demonstrate potential directions for further work:

- Incorporate more diverse datasets including emerging health topics, regional content and multilingual sources to improve model accuracy and generalization.
- Extend the system to support additional languages, enabling a broader user base to access reliable health information and ensuring inclusivity for non-English speakers.
- Allow users to provide feedback on verification results to enhance model accuracy, user trust and system transparency.
- Improve the clarity of verification outputs by providing simple, user-friendly explanations based on the model's reasoning.
- Introduce an explainable AI system (XAI) to support more reasoning for the claim uploaded by user
- Ensure user data and queries are securely stored and processed with strong encryption and privacy controls to maintain trust and comply with data protection regulations.

In summary, the system provides a basic foundation for tackling online health misinformation. Implementing these further works and recommendations will improve its accuracy, usability and reach. Ultimately this enables users to make safer, informed decisions regarding their health.

10 PROJECT TASK AND TIME SCHEDULE

The project schedule has been designed as per the requirements involved. The project is estimated to be completed in about 10-12 weeks. Research and requirement analysis is to be done first and is also crucial for the overall working of the project explaining the lengthy time requirements. The project will be well documented on all the iterations reporting the working of the project at each time. Debugging and testing are to be done prior to the completion of the project.

TASK	APPROX DURATION IN DAYS
Requirement Analysis, Base AI Prototype	13
OCR Plugin & UX Research	10
Mobile App Prototype, Harvesting Model, & Backend Setup	8
AI Detection Model, Indexing Model	12
UI/UX Design & System Integration & Backend& Core Mobile App	10
Testing and Bug Fixing	14

Table 3: Time Schedule

10.1 Gantt Chart

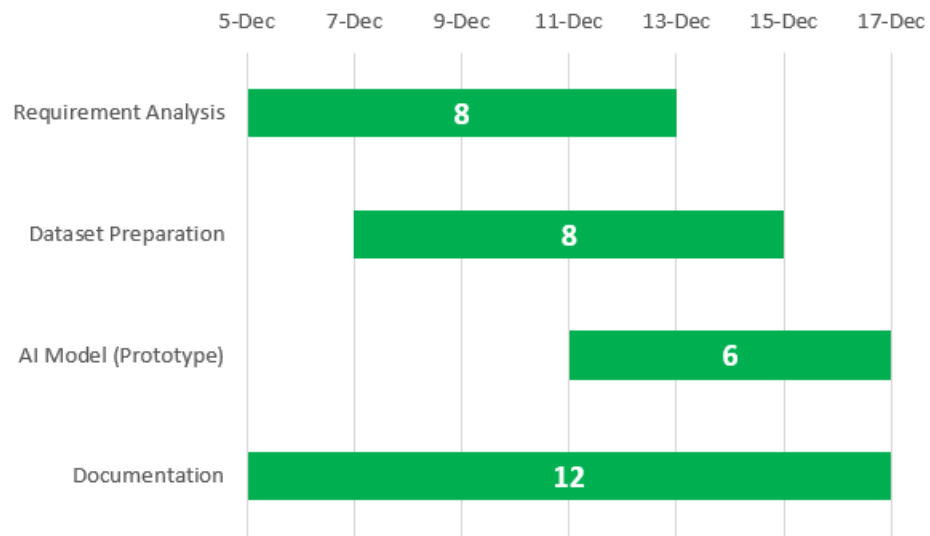


Figure 30: Increment 1

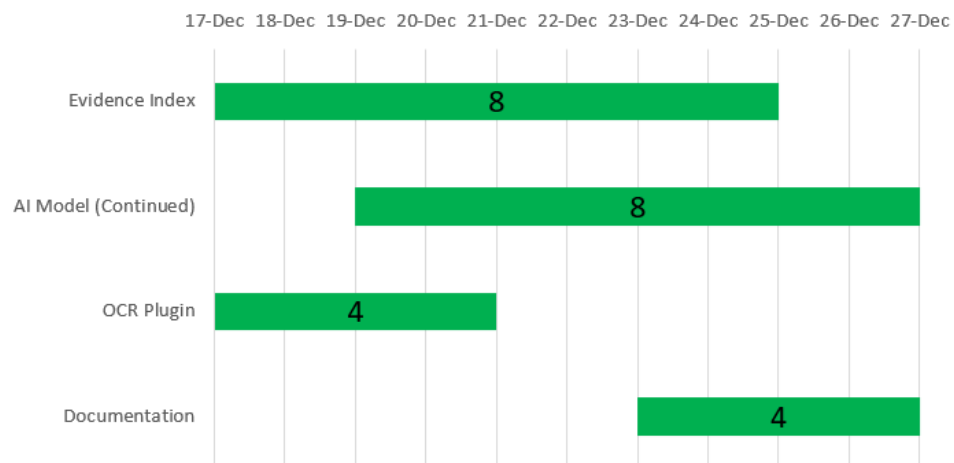


Figure 31: Increment 2

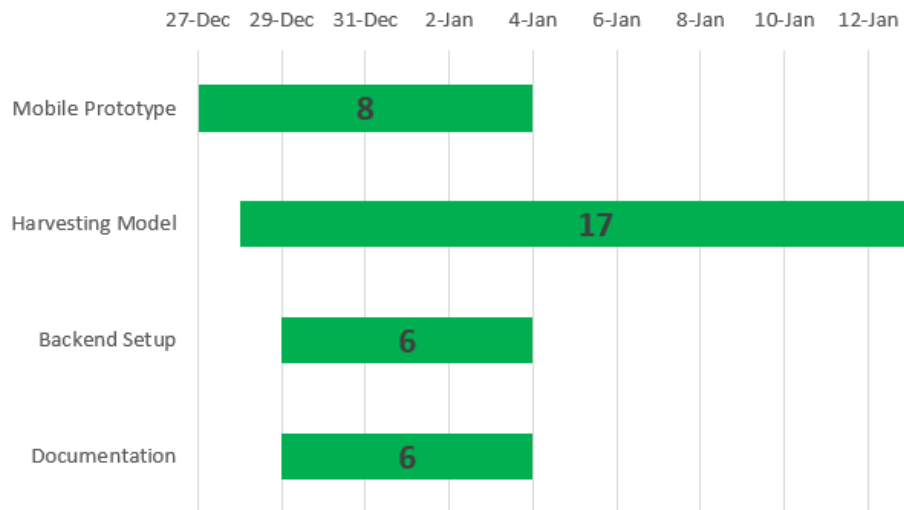


Figure 32: Increment 3

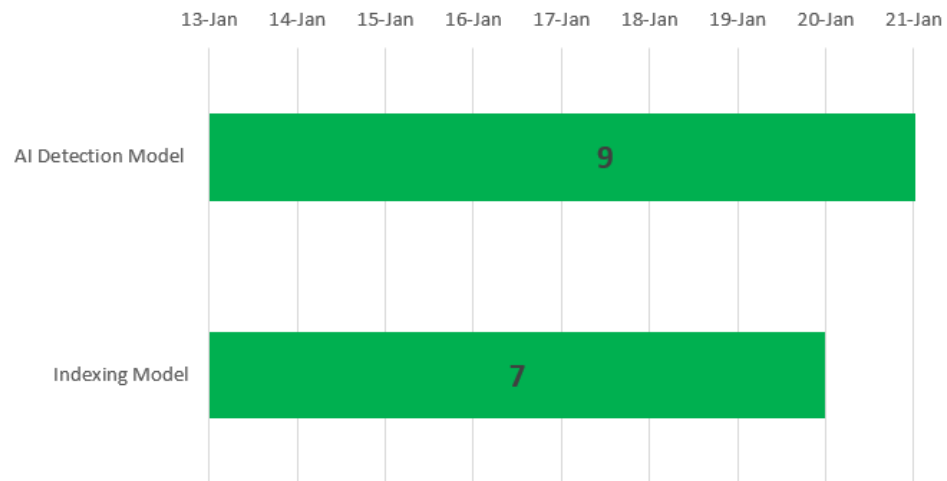


Figure 33: Increment 4

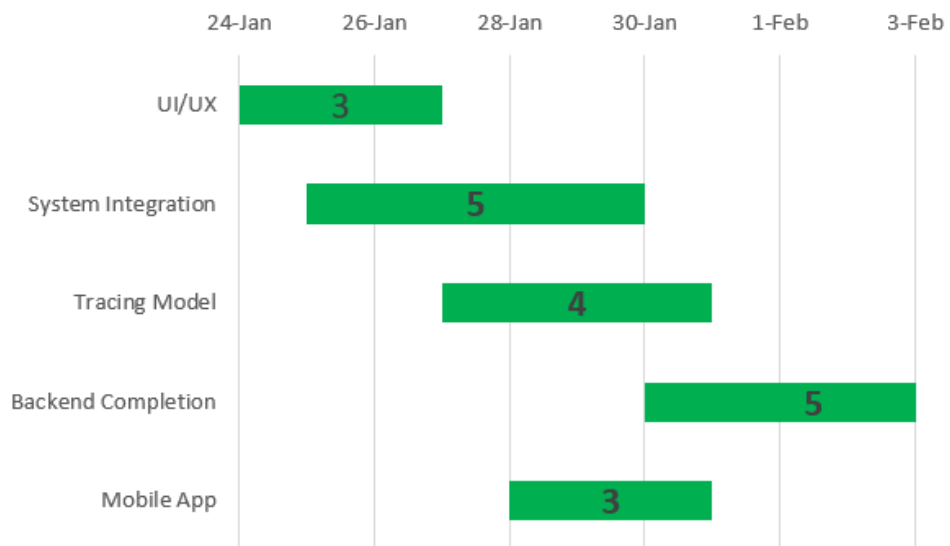


Figure 34: Increment 5

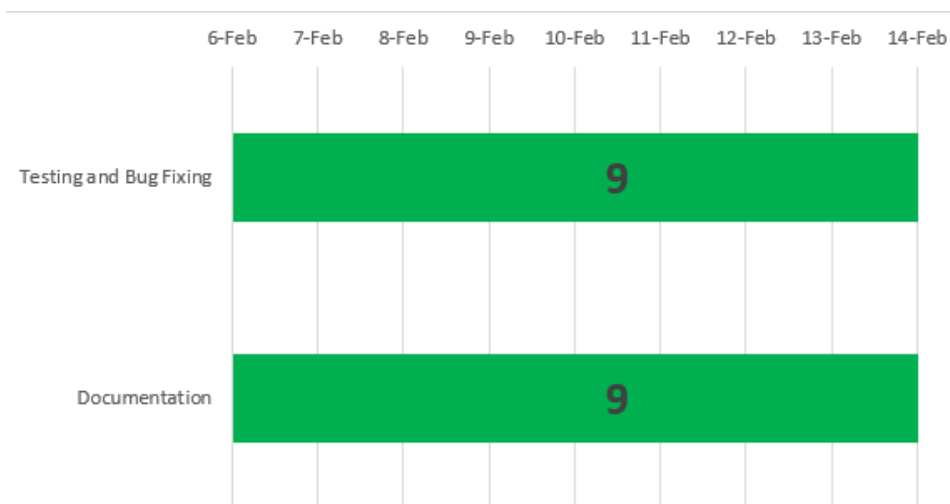


Figure 35: Increment 6

10.2 Task Division

TEAM MEMBER	ROLE
Aman Sheikh	Mobile App, Data Harvesting, QA System and Research
Prashant Rokaya	Backend and Database
Shikshya K.C.	UX Research, UI Design and Documentation
Shreya Khanal	AI Model and Research

Table 4: Division of roles of team members

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Appendix



Your Answer

Conclusion

Sleeping more than 8 hours may be associated with increased risk of heart disease, stroke, and death, as suggested by the European Heart Journal study [Source 3].

Evidence

The study found that those who slept more than 10 hours per day had a 41% greater risk of these outcomes.

Sources



How to Recognize and Manage Dysania



<https://www.webmd.com/mental-health/what-is-dysania>

Low activity and too much sleep can also be bad for your health. The recommended amount of sleep for most adults is 6 to 8 hours per night. A 2014 study ...

Score 0.89



How Sleep Affects Your Cholesterol Levels



<https://www.webmd.com/cholesterol-management/how-...>

Oversleeping can affect cholesterol, too. In one

QA System X

ng more than 8 hours good for health?



Your Results



Dengue – the Region of the Americas



By WHO

Dengue is a mosquito - borne viral disease (arbovirus) that has the potential to cause serious public health impact. The virus that causes this in...

Score 0.96



Dengue Fever



By Stephanie Watson

Dengue (pronounced DEN-gee) fever is a mosquito-borne disease caused by any one of four closely related dengue viruses. You get dengue fe...

Score 0.91



Dengue - Iran (Islamic Republic of)



By WHO

Dengue is a mosquito-borne viral disease caused by the dengue virus (DENV), with the potential to cause a serious public health impact. Dengue infe...

Score 0.90

Doc Search X

Q Dengue due to mosquito



Home



History

Your Results



Rabies - Timor-Leste



By WHO

On 30 May 2025, a second fatal case of rabies was confirmed in an adult male from Oecusse Municipality. The person had been bitten by a dog ...

Score 0.95



What Veterans Should Know About Rabies



By Taylor Wendt

Rabies infection can quickly become life-threatening if you've been bitten by a rabid animal and their saliva makes contact with a scratch or o...

Score 0.92



Rabies: Symptoms and Treatment



By Abby Alten Schwartz, WebMD Editorial Contributor, Kristin Mitchell

The rabies virus causes the rabies infection and can spread through saliva, often from an infected animal biting another animal or human.

Score 0.92

Doc Search X

🔍 Rabies Infection



Home



History



Analysis Result



⊗ REFUTES

View Evidence

Extracted Text

Ginger curse cancer.

Scientific Evidence



Home Remedies for Ear Pain



<https://www.webmd.com/cold-and-flu/ear-infection/ear-...>

Ginger

Score 0.46



Antiemetics for Migraine With Nausea



<https://www.webmd.com/migraines-headaches/migraine-...>

Ginger. Studies on pregnant women dealing with morning sickness and people with nausea from chemotherapy show that it is an effective and inex...

Score 0.41



Aristolochic Acids and Bladder Cancer



[https://www.webmd.com/cancer/bladder-cancer/aristol-](https://www.webmd.com/cancer/bladder-cancer/aristol-...)



Analysis Result



✔ SUPPORTS

View Evidence

Extracted Text

Sleeping 6 to 8 hours is good for mental wellbeing.

Scientific Evidence



How to Recognize and Manage Dysania

<https://www.webmd.com/mental-health/what-is-dysania>

Low activity and too much sleep can also be bad for your health. The recommended amount of sleep for most adults is 6 to 8 hours per night. A 2014 study ...

Score 0.88



Why Can't I Sleep?

<https://www.webmd.com/sleep-disorders/cm/why-cant-i...>

Although sleep experts agree that most adults need 7-8 hours of restorative sleep per night for optimal wellness, not everyone is the same. Some ...

Score 0.85

Good morning

How can I help you fact-check today?

 QA System

 Doc Search

 Scan Image

 Upload Image

 Enter your Claim...



History

✓ All

Verifier

Q&A

Retriever

Today



Sleeping 6 to 8 hours is good for mental wellbeing.

10:23

supports



Ginger curse cancer.

10:22

refutes



Guner curse cancer.

10:22



Exercising is cures cancer.

10:22

supports



Exercising is bad for you.

10:22

Not Enough Evidence



Sugary drink doesn't cause spik...

10:22

Not Enough Evidence



Sugary drink causes spike in sugar levels.

10:21

refutes



Home



History

History

- All
- ✓ Verifier
- Q&A
- Retriever

Today

- 

Sleeping 6 to 8 hours is good for mental wellbeing.

10:23

supports
- 

Ginger curse cancer.

10:22

refutes
- 

Guner curse cancer.

10:22
- 

Exercising is cures cancer.

10:22

supports
- 

Exercising is bad for you.

10:22

Not Enough Evidence
- 

Sugary drink doesn't cause spik...

10:22

Not Enough Evidence
- 

Sugary drink causes spike in sugar levels.

10:21

refutes



History

- All
- Verifier
- ✓ Q&A
- Retriever

Today

-  Do mosquito cause cancer
10:09
-  is it actually true that lemon
cures cancer
09:56
-  does ginger cures cancer?
09:49
-  Sugary drink cause diabetes
09:26
-  does smoke cause cancer
09:25
-  does smoke cause camcer
09:25
-  Does smoke kill mosquitoes?
09:19



History

- All
- Verifier
- Q&A
- ✓ Retriever

Today



Rabies Infection
10:20



exercise leads to weight loss
09:53



does smoke kills mosquitoes
09:21



abc xyz
09:15



abc xyz
09:15



abc xyz
09:15



Dengue due to mosquito
09:11

